

A
PROJECT REPORT
(PS04CAST55)
ON

**“DATA-DRIVEN EVALUATION OF FMCG PRODUCT
PERFORMANCE”**

A
PROJECT REPORT
SUBMITTED IN PARTIAL
FULFILMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE OF
M.Sc. in Applied Statistics

SUBMITTED BY
CHAUHAN TUSHARBHAI SURESHBHAI
(ROLL NO: 16)

UNDER THE GUIDANCE OF
Mr. VIRAJ M. TUR

**DEPARTMENT OF STATISTICS
SARDAR PATEL UNIVERSITY,
VALLABH VIDYANAGAR – 388120,
ANAND, GUJARAT**

April-2026



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CERTIFICATE

This is to certify that MR. CHAUHAN TUSHARBHAI SURESHBHAI, Roll No. 16, student of Master of Science in Applied Statistics, has satisfactorily completed his project work on “DATA-DRIVEN EVALUATION OF FMCG PRODUCT PERFORMANCE” under the Paper Project Work (PS04CAST55) for M.Sc. Applied Statistics Semester IV during the period NOV 2025 – APRIL 2026.

Place: Department of Statistics, V.V Nagar, Anand

Date: 6th April 2026

Mr. Viraj M. Tur
(Project Guide)

Prof. (Dr.) Dharmesh Raykundaliya
(Head of Department)

DECLARATION

I hereby declare that the report entitled “**DATA-DRIVEN EVALUATION OF FMCG PRODUCT PERFORMANCE**” is an original work carried out by me, **Chauhan Tusharbhai Sureshbhai**, under the guidance and supervision of **Mr. Viraj M. Tur**. This work has not been submitted, either in full or in part, to this or any other University for the award of any degree, diploma, or any other academic qualification.

I further declare that all the sources of information used in this study have been properly acknowledged.

Place: Department of Statistics, V.V Nagar, Anand

Date: 6th April 2026

Signature

Chauhan Tusharbhai Sureshbhai

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Chauhan Tusharbhaj Sureshbhai

Place: Dept. of Statistics, V.V Nagar, Anand

Date: 6th April 2026

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Abstract

This study analyzes sales data from the FMCG sector to understand product performance and support data-driven decision-making. The analysis focuses on key objectives including sales performance, the impact of price on sales, and the effect of promotional activities. Using a structured dataset with variables such as units sold, pricing, stock availability, and promotions, descriptive statistics and non-parametric methods were applied due to the presence of skewness and non-normality. A Negative Binomial regression model was used to assess the impact of promotions on sales, as it is suitable for overdispersed count data. In addition, the study examines channel performance and evaluates inventory and stock efficiency to identify variations across channels and detect issues such as overstocking and stock-outs. Correlation analysis and visualization techniques were used to explore relationships between stock levels and sales. Finally, a LightGBM model with time-based and lag features was developed to predict sales for the last 42 days. The results highlight the usefulness of statistical and machine learning methods in understanding sales behaviour and improving overall operational efficiency.

1. INTRODUCTION

1.1 Background of the Study

The fast-moving consumer goods (FMCG) sector is characterized by high sales volume, rapid inventory turnover, and intense market competition. In such an environment, understanding product performance is essential for businesses to maintain efficiency and meet consumer demand effectively. Sales of FMCG products are influenced by multiple factors such as pricing, promotional activities, distribution channels, and stock availability. Analyzing these factors helps organizations gain insights into consumer behavior and operational performance.

With the increasing availability of data, businesses are shifting towards data-driven approaches to evaluate and improve their performance. Sales data provides valuable information that can be used to identify trends, measure the effectiveness of promotions, and understand how different variables impact demand. However, real-world sales data often exhibits challenges such as skewness, non-normal distribution, and variability across regions and channels, which require appropriate statistical techniques for accurate analysis. This study focuses on analyzing FMCG sales data to examine key aspects such as sales performance, price impact, promotional effectiveness, channel differences, and inventory efficiency. In addition, predictive modeling is used to forecast future sales based on historical patterns. By applying statistical and machine learning methods, the study aims to provide meaningful insights into sales behavior and operational efficiency, supporting better understanding of product performance in the FMCG sector.

1.2 Problem Statement

The FMCG sector operates in a highly dynamic environment where sales performance is influenced by multiple factors such as pricing, promotional activities, distribution channels, and inventory availability. Despite the availability of large volumes of sales data, many businesses face challenges in effectively analyzing this data to understand product performance and demand patterns. In particular, issues such as fluctuating sales, inefficient inventory management, and varying performance across channels and regions make it difficult to make informed decisions.

Additionally, real-world sales data often exhibits non-normality, skewness, and overdispersion, which limit the effectiveness of traditional analytical methods. There is a need to apply appropriate statistical and analytical techniques to accurately assess the impact of key factors such as price and promotions on sales. Furthermore, businesses require reliable methods to forecast future sales in order to better plan inventory and operations.

Therefore, this study aims to address these challenges by using data-driven approaches to evaluate product performance, analyze key influencing factors, and develop a predictive model for short-term sales forecasting.

1.3 Objectives of the Study

Objective 1: Sales Performance Analysis

To analyze overall sales patterns across time (year, month, day) and evaluate performance across regions, channels, categories, brands, and pack types using descriptive and visual techniques.

Objective 2: Price Impact on Sales

To examine the relationship between price per unit and units sold using correlation analysis and statistical tests to determine whether pricing significantly influences sales.

Objective 3: Effect of Promotions on Sales

To evaluate the impact of promotional activities on sales and revenue using comparative analysis and non-parametric tests, along with modeling using a Negative Binomial approach.

Objective 4: Channel Performance Comparison

To compare sales and revenue performance across different distribution channels and assess whether significant differences exist using statistical testing methods.

Objective 5: Inventory & Stock Efficiency

To analyze the relationship between stock availability and sales, and evaluate inventory efficiency using metrics such as stock-to-sales ratio and inventory turnover.

Objective 6: Sales Prediction (42-Day Forecasting)

To develop a machine learning model using time-based and lag features to predict future sales for a 42-day period and evaluate model performance using accuracy metrics.

1.4 Scope of the Study

This study focuses on analyzing FMCG sales data to understand product performance across multiple dimensions such as time, region, channel, category, and brand. The analysis is based on historical data covering the period from 2022 to 2024 and includes key variables such as units sold, price, stock availability, promotions, and revenue. The study applies descriptive analysis, statistical techniques, and data visualization methods to identify patterns and relationships in the data.

The scope also includes evaluating the effect of promotions and inventory efficiency using appropriate statistical models, including the Negative Binomial model for count-based sales data. Additionally, a machine learning approach is used to forecast sales for the next 42 days using time-based and lag features. However, the study is limited to the available dataset and does not account for external factors such as market competition, seasonal events, or consumer behaviour changes beyond the given data.

1.5 Limitations of the Study

This study is based on secondary data obtained from Kaggle, which is simulated in nature. As a result, the dataset may not fully reflect real-world market conditions, customer behaviour, or business complexities, limiting the practical applicability of the findings. Since the data is artificially generated, certain relationships (such as price impact on sales) may appear weak or unrealistic compared to actual industry scenarios.

Additionally, the analysis is restricted to the variables available in the dataset and does not include external factors such as competitor actions, economic conditions, seasonal events, or consumer preferences, which can significantly influence sales. The predictive models are also dependent on historical patterns within the dataset, and their performance may vary when applied to real-world data. Therefore, the results should be interpreted with caution and considered primarily for academic and analytical purposes.

2. LITERATURE REVIEW

2.1 Overview of Previous Studies

Previous studies highlight the importance of data-driven techniques in analyzing sales performance, identifying trends, and understanding product demand across time, regions, and categories. Descriptive analysis and statistical methods are commonly used to evaluate patterns and support decision-making.

Research on pricing shows mixed results, where some studies find a significant impact on sales, while others suggest weak relationships depending on product and market conditions. Promotional activities are generally found to have a strong positive effect on sales and revenue, often analyzed using hypothesis testing and count-based models such as the Negative Binomial model.

Inventory management studies emphasize the role of efficient stock control using metrics like inventory turnover and stock-to-sales ratio. Recent research also focuses on machine learning models for sales forecasting, which use time-based and lag features to improve prediction accuracy.

2.2 Methods Used by Other Authors

Previous studies have applied a combination of statistical and analytical methods to evaluate sales performance and demand patterns. Descriptive statistics and data visualization techniques are commonly used to summarize data and identify trends across time, regions, and product categories.

Correlation analysis and non-parametric tests such as Spearman correlation, Mann–Whitney U test, and Kruskal–Wallis test are widely used to examine relationships and compare groups when data does not follow normal distribution. For count-based sales data, regression models such as Poisson and Negative Binomial models are frequently applied to handle overdispersion and model sales behavior effectively.

In recent studies, machine learning techniques such as regression-based models and tree-based algorithms have been used for sales prediction. These models often incorporate time-based features, lag variables, and rolling statistics to capture patterns and improve forecasting accuracy.

2.3 Research Gap

Although previous studies have extensively analyzed sales performance, pricing effects, promotions, and inventory management, many of them focus on individual factors rather than providing a comprehensive analysis that combines all key aspects in a single framework. There is limited integration of sales, promotions, channel performance, and inventory efficiency in one study.

Additionally, while machine learning models are increasingly used for sales forecasting, fewer studies combine statistical modeling approaches (such as the Negative Binomial model) with advanced predictive techniques on the same dataset. Another gap is the lack of studies using simulated datasets to test model performance and validate analytical approaches.

3. DATA DESCRIPTION

3.1 Source of Data

The data used in this study is obtained from Kaggle, an online platform that provides datasets for research and analytical purposes. The dataset is secondary in nature and is synthetically generated to simulate real-world Fast Moving Consumer Goods (FMCG) sales transactions under practical business conditions.

This dataset represents daily-level sales data over a period of three years (2022–2024). It is inspired by real-world FMCG companies and is designed to support analysis in areas such as demand forecasting, promotion effectiveness, and sales performance evaluation. The data includes a structured product hierarchy (SKU, brand, segment, and category), along with multiple business dimensions such as sales channels (Retail, Discount, and E-commerce) and regions located in Poland (Central, North, and South).

The dataset contains key variables including date, units sold, price per unit, revenue, promotion status, stock availability, delivery time, and pack type (Single, Multipack, Carton). It also incorporates realistic patterns such as seasonal trends, gradual price changes over time, and the introduction of new products in later periods. These characteristics make the dataset suitable for applying statistical analysis and machine learning techniques in a controlled yet practical environment.

Dataset Link - <https://www.kaggle.com/datasets/beatafaron/fmcg-daily-sales-data-to-2022-2024>

3.2 Variable Explanation

Variable Name	Description
Date	Transaction date of sales
Unique_ID (SKU)	Unique product identifier (e.g., MI-006)
Brand	Brand name of the product
Segment	Sub-category within a product category
Category	Main product category (Milk, Yogurt, SnackBar, etc.)
Channel	Sales channel (Retail, Discount, E-commerce)
Region	Geographic region (PL-Central, PL-North, PL-South)
Pack_Type	Packaging type (Single, Multipack, Carton)
Price_Unit	Price per unit of the product
Promotion_Flag	Indicates promotion (1 = Yes, 0 = No)
Delivery_Days	Number of days required for delivery
Stock_Available	Available inventory level
Delivered_Qty	Quantity delivered
Units_Sold	Number of units sold (dependent variable)
Revenue	Total sales value (Units Sold $\times$ Price per Unit)
Year	Extracted year from date
Month	Extracted month from date
Day	Extracted day from date
Price_Range	Price category (Low, Medium, High)
Stock_to_Sales_Ratio	Ratio of stock available to units sold
Inventory_Turnover	Ratio of units sold to stock available

Table 3.1: Variable Explanation

3.3 Sample Size

The dataset used in this study consists of **190,757 observations (rows)** and **14 variables (columns)**. Each observation represents a transaction or record related to FMCG product sales across different time periods, regions, and channels.

This large sample size provides sufficient data for conducting reliable statistical analysis, modeling, and forecasting, ensuring better accuracy and robustness of the results.

3.4 Data Collection Method

The data used in this study was collected from Kaggle, an online data repository, and is secondary in nature. The dataset was downloaded directly from the platform and does not involve any primary data collection methods such as surveys or interviews.

Since the data is simulated, it was originally generated to represent real-world FMCG sales scenarios, including various attributes like sales, pricing, promotions and inventory. After collection, the data was cleaned, preprocessed, and structured for analysis, including handling missing values, creating new features and preparing it for statistical and machine learning models.

4. METHODOLOGY

4.1 Descriptive Statistics

Descriptive statistics were used to summarize and understand the basic characteristics of the dataset. Key measures such as **mean, median, standard deviation, minimum, maximum, and quartiles** were calculated for important numerical variables including *price_unit*, *units_sold*, *stock_available*, *delivered_qty*, and *revenue*.

These statistical measures help in identifying the **central tendency, dispersion, and overall distribution** of the data, providing a clear overview of sales behaviour and variability across different variables.

4.1.1 Mathematical Representation:

- **Mean:**

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

- **Variance:**

$$\sigma^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$$

- **Standard Deviation:**

$$\sigma = \sqrt{\sigma^2}$$

Overall, descriptive statistics form the foundation of the analysis by helping to understand data patterns and supporting further statistical and modelling techniques.

- Mean → shows the average value of the variable
- Median → shows the middle value, useful for skewed data
- Mode → shows the most frequent value
- Standard Deviation and Variance → measure the spread of data
- Skewness → indicates whether the data is symmetric or skewed
- Kurtosis → indicates the shape and peak of the distribution

These measures help in understanding whether the data is normally distributed or not.

4.2 Graphical Analysis

Graphical analysis was performed to visually explore the dataset and understand patterns, trends, and relationships among variables. Various visualization techniques such as **bar charts, line plots, histograms, boxplots, pie charts, and heatmaps** were used.

- Bar Charts and Pie Charts:

These were used to analyze sales distribution across categories, brands, regions, channels, and pack types. They help in identifying top-performing segments and understanding contribution levels.

- Line Plots:

Line charts were used for time-based analysis (year-wise, month-wise, and day-wise trends) to observe patterns such as growth, seasonality, and stability in sales.

- Histograms and Density Plots:

These were used to examine the distribution of numerical variables and identify skewness, concentration of values, and overall data spread.

- Boxplots:

Boxplots were used to detect outliers and compare distributions across groups such as price ranges, channels, and promotion status.

- Scatter Plots:

Scatter plots were used to analyze relationships between variables (e.g., price vs units sold, stock vs sales), helping to identify correlation patterns.

- Heatmaps:

Heatmaps were used to visualize correlations between variables and to analyze category-wise sales across months, highlighting relationships and seasonal patterns.

Overall, graphical analysis provided clear and intuitive insights into data behaviour, supporting better interpretation and validation of statistical results.

4.3 Correlation Analysis

Correlation analysis was conducted to examine the relationship between key variables in the dataset, particularly to understand how factors such as **price, promotions, stock availability, and inventory metrics** influence sales.

Two types of correlation methods were used:

- **Pearson Correlation:**
Used to measure the **linear relationship** between continuous variables.
- **Spearman Correlation:**
Used to measure the **monotonic (rank-based) relationship**, especially suitable for non-normal data.

Mathematical Representation (Pearson Correlation):

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sigma_X \sigma_Y}$$

Where:

- r = correlation coefficient
- X, Y = variables
- $\bar{X}, \bar{Y}$ = means
- σ_X, σ_Y = standard deviations

The correlation coefficient ranges from **-1 to +1**:

- **+1** → Strong positive relationship
- **-1** → Strong negative relationship
- **0** → No relationship

Correlation analysis helps in identifying **important relationships between variables**, supporting better feature selection and understanding of factors affecting sales performance.

4.4 Spearman's Rank Correlation Test

Spearman's Rank Correlation Test was used to measure the **strength and direction of the monotonic relationship** between two variables. Unlike Pearson correlation, it does not require the data to be normally distributed and works based on **ranked values**.

It is particularly suitable for this study because the data shows **non-normality and skewness**.

Mathematical Representation:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

Where:

- ρ = Spearman correlation coefficient
- d_i = difference between ranks of each observation
- n = number of observations

The value of ρ ranges from **-1 to +1**, where:

- **+1** indicates a perfect positive monotonic relationship
- **-1** indicates a perfect negative monotonic relationship
- **0** indicates no relationship

Assumptions of Spearman's Rank Correlation Test

- **Ordinal or Continuous Data:**
Variables should be measured on at least an ordinal scale (ranking should be possible).
- **Monotonic Relationship:**
The relationship between variables should be monotonic (consistently increasing or decreasing), though not necessarily linear.
- **Paired Observations:**
Each observation in one variable must correspond to an observation in the other variable.
- **Independence of Observations:**
Each pair of observations should be independent of others.
- **No Requirement of Normality:**
The data does not need to follow a normal distribution.
- **Robust to Outliers:**
Since the test is based on ranks, it is less affected by extreme values.

Overall, Spearman's test is a reliable method for analyzing relationships in **non-normal and skewed datasets**, making it suitable for this study.

4.5 Mann–Whitney U Test

The Mann–Whitney U Test is a **non-parametric statistical test** used to compare whether there is a **significant difference between two independent groups**. It is an alternative to the independent t-test when the data does not follow a normal distribution.

In this study, the test is used to compare **units sold between promoted and non-promoted products**.

Purpose of the Test

- To determine whether two independent groups come from the same distribution
- To check if one group tends to have higher values than the other

Hypotheses

- Null Hypothesis (H_0):
There is **no significant difference** in units sold between promoted and non-promoted products.
- Alternative Hypothesis (H_1):
There is a **significant difference** in units sold between the two groups.

Mathematical Concept

The test ranks all observations from both groups together and compares the **sum of ranks**.

$$U = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1$$

Where:

- n_1, n_2 = sample sizes
- R_1 = sum of ranks of group 1

Assumptions of Mann–Whitney U Test

- Independent Groups:
The two groups must be independent (e.g., promoted vs non-promoted).
- Ordinal or Continuous Data:
The dependent variable should be at least ordinal.
- Independent Observations:
Each observation should be independent of others.
- Similar Shape of Distributions (Optional):
The distributions of both groups should have a similar shape for median comparison.
- No Normality Requirement:
The test does not require normally distributed data.

Interpretation in Study Context

The Mann–Whitney U Test helps determine whether **promotions significantly affect sales performance**. Since the data is not normally distributed, this test provides a reliable way to compare the two groups.

This test is particularly suitable for this study as it handles **skewed data and outliers effectively**, making it ideal for real-world sales data analysis.

4.5 Kruskal–Wallis Test

The Kruskal–Wallis Test is a **non-parametric statistical test** used to compare **three or more independent groups** to determine whether there is a significant difference between them. It is an extension of the Mann–Whitney U Test and serves as a non-parametric alternative to one-way ANOVA.

In this study, it is used to compare variables such as **units sold across price groups or sales channels**.

Purpose of the Test

- To check whether **multiple groups differ significantly**

- To determine if at least one group has a different median

Hypotheses

- Null Hypothesis (H₀):
All groups have the **same distribution (no significant difference in medians)**.
- Alternative Hypothesis (H₁):
At least **one group differs significantly** from the others.

Mathematical Representation

$$H = \frac{12}{n(n+1)} \sum \frac{R_i^2}{n_i} - 3(n+1)$$

Where:

- H = Kruskal–Wallis test statistic
- n = total number of observations
- R_i = sum of ranks for group i
- n_i = number of observations in group i

Assumptions of Kruskal–Wallis Test

- Independent Groups:
The groups being compared must be independent.
- Ordinal or Continuous Data:
The dependent variable should be at least ordinal.
- Independent Observations:
Each observation must be independent.
- Similar Distribution Shape (Optional):
Groups should have a similar distribution shape for median comparison.
- No Normality Requirement:
The test does not require normally distributed data.

Interpretation in Study Context

The Kruskal–Wallis Test is useful for analyzing whether factors such as **price levels or sales channels** significantly influence units sold. It is appropriate for this study because the data does not strictly follow normal distribution and may contain outliers.

This test provides a robust method for comparing multiple groups in **real-world sales data**, where traditional parametric assumptions may not hold.

4.6 Inventory Efficiency Metrics

Inventory efficiency metrics are used to evaluate how effectively inventory is managed in relation to sales. These metrics help in understanding whether stock levels are optimal or excessive.

(I) Stock-to-Sales Ratio

The **Stock-to-Sales Ratio** measures how much inventory is available compared to the number of units sold.

Formula:

$$\text{Stock-to-Sales Ratio} = \frac{\text{Stock Available}}{\text{Units Sold}}$$

Interpretation:

- A **higher ratio** indicates excess inventory or slow-moving stock
- A **lower ratio** indicates efficient inventory usage or high demand

In this study, the average stock-to-sales ratio suggests that inventory levels are relatively high compared to sales, indicating potential overstocking.

(II) Inventory Turnover Ratio

The **Inventory Turnover Ratio** measures how efficiently inventory is converted into sales.

Formula:

$$\text{Inventory Turnover} = \frac{\text{Units Sold}}{\text{Stock Available}}$$

Interpretation:

- A **higher turnover ratio** indicates efficient inventory management and faster sales
- A **lower turnover ratio** indicates slower movement of stock

In this study, the turnover ratio is relatively low, suggesting that inventory is not being converted into sales quickly.

Overall Insight

Together, these metrics indicate that while stock availability is sufficient, **inventory utilization can be improved** by better aligning stock levels with actual sales demand.

4.7 Regression Modelling (Negative Binomial Model)

The Negative Binomial Regression Model was used to analyze the relationship between units sold (dependent variable) and key predictors such as promotion_flag and stock_available.

This model is suitable because:

- The dependent variable (**units_sold**) is **count data**
- The data shows **overdispersion** (variance > mean)
- Poisson regression assumptions are violated

Model Specification

$$\log(\mu_i) = \beta_0 + \beta_1(\text{promotion_flag}) + \beta_2(\text{stock_available})$$

Where:

- μ_i = expected units sold
- β_0 = intercept
- β_1, β_2 = coefficients of predictors

Estimation Method

The model parameters are estimated using **Maximum Likelihood Estimation (MLE)**, which finds the values of coefficients that maximize the likelihood of observing the given data.

Key Assumptions

- **Count Dependent Variable:**
The response variable should be count data (non-negative integers)
- **Overdispersion Present:**
Variance of the dependent variable is greater than its mean
- **Independence of Observations:**
Each observation is independent
- **Log-linear Relationship:**
Predictors have a linear relationship with the log of expected counts

Interpretation of Coefficients

- Coefficients are in log form, so they are interpreted using exponentiation:

$$e^{\beta}$$

- If $e^{\beta} > 1$: positive effect on sales
- If $e^{\beta} < 1$: negative effect on sales

Relevance to Study

The Negative Binomial model is appropriate for this study as it effectively handles **overdispersed sales data** and provides reliable insights into how factors like **promotion and stock availability influence sales performance**.

4.8 Model Evaluation

The performance of the predictive model was evaluated using multiple metrics to assess its **accuracy, reliability, and goodness of fit**.

4.8.1 RMSE (Root Mean Squared Error)

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

RMSE measures the **average magnitude of prediction errors**, giving higher weight to larger errors. A **lower RMSE** indicates better model performance.

4.8.2 MAE (Mean Absolute Error)

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

MAE represents the **average absolute difference** between actual and predicted values. It is easy to interpret and less sensitive to outliers compared to RMSE.

4.8.3 MAPE (Mean Absolute Percentage Error)

$$MAPE = \frac{1}{n} \sum \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100$$

MAPE expresses the error as a **percentage**, making it useful for understanding prediction accuracy in relative terms. Lower MAPE indicates better predictive accuracy.

4.8.4 R² (Coefficient of Determination)

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

R² measures how well the model explains the **variation in the dependent variable**. Values closer to **1 indicate a better fit**.

Overall Purpose

These metrics collectively help evaluate how well the model captures patterns in the data and how accurately it predicts sales, ensuring the model is **robust and reliable for decision-making**.

4.9 Time-Based Modelling (Lag Features)

To capture the **time dependency in sales data**, a lag variable (**lag_1**) was created, representing the previous period's sales.

$$\text{lag_1} = \text{units_sold}_{t-1}$$

This means that the sales value from the **previous time period (t-1)** is used as a predictor for the current period.

Purpose of Lag Feature

- Captures **temporal patterns** in sales
- Helps model **sales continuity and trends**
- Reflects the idea that **past sales influence current sales**

Importance in This Study

Including the lag variable improves the model by incorporating **historical information**, making predictions more accurate and realistic. It shows that sales are not random but are **partially dependent on previous periods**.

4.10 Machine Learning Model (LightGBM)

A **Light Gradient Boosting Machine (LightGBM)** model was applied for forecasting sales, as it is well-suited for handling large-scale and complex datasets.

LightGBM is an **ensemble learning algorithm** based on decision trees, which builds multiple trees sequentially to improve prediction accuracy.

Key Advantages

- Efficient for Large Datasets:
Optimized for faster training and lower memory usage
- Captures Non-linear Relationships:
Effectively models complex patterns between variables

- High Predictive Performance:
Uses boosting technique to minimize errors iteratively

Model Features Used

The model incorporates multiple types of features:

- Time-based Features:
Year, Month, and Day to capture seasonal patterns
- Lag Features:
Previous sales values (e.g., lag_1) to include time dependency
- Inventory Variables:
Stock availability and related metrics
- Promotion Variables:
Promotion_flag to capture the effect of promotions

Relevance to Study

LightGBM enhances forecasting accuracy by combining **historical patterns, operational factors, and temporal trends**, making it a powerful tool for predicting future sales in a dynamic FMCG environment.

4.11 Forecasting Approach

A **time-based train-test split** was used to ensure realistic and reliable forecasting.

- Training Data:
Historical data was used to train the model and learn patterns.
- Testing Data:
The **last 42 days** of data were kept separate to evaluate model performance.

Purpose of This Approach

- Maintains the **chronological order** of data
- Prevents **data leakage** (future data influencing past predictions)
- Reflects **real-world forecasting scenarios**

Importance in This Study

This approach ensures that the model is evaluated on **unseen future data**, making the predictions more **practical, accurate, and reliable** for real-world decision-making.

5. DATA PREPROCESSING

5.1 Data Cleaning

5.1.1 Handling Missing Values:

The dataset was checked for missing values using data validation techniques, and no missing values were found across all variables, including date, product details, sales, pricing, promotion status, and inventory fields.

5.1.2 Data Type Conversion:

The dataset's data types were carefully examined to ensure all variables were in appropriate formats for analysis. It was observed that the date column was initially stored in object format, which was then converted into datetime format. This conversion improves data handling, enables time-based analysis, and ensures accurate results in further processing.

5.1.3 Handling Invalid Values:

To ensure data quality and reliability, the dataset was examined for invalid or inconsistent values. A check on numerical variables revealed the presence of negative values in **stock_available**, **delivered_qty**, and **units_sold**, which are logically invalid in the context of sales data.

A total of **3 negative observations** were identified in each of these variables. Since such values can distort analysis and model performance, these records were removed from the dataset.

After cleaning, the dataset size was reduced from **190,757 to 190,754 observations**. The removal of only 3 records is negligible compared to the overall dataset size and does not significantly impact the analysis or results, ensuring the robustness of the study.

5.1.4 Duplicate Records:

The dataset was checked for duplicate entries, and no duplicate records were found. Therefore, no data removal or modification was required in this step.

5.2 Data Transformation

5.2.1 Feature Engineering:

Feature engineering was performed to enhance the dataset by creating meaningful variables for analysis. Date-related features such as **Year, Month and Day** were extracted from the date column to capture temporal patterns. Additionally, the **price_unit** variable was categorized into three groups (Low, Medium, High) to analyze sales behavior across different price levels.

Further, inventory-related metrics were developed, including the **stock-to-sales ratio** and **inventory turnover ratio**, to evaluate stock efficiency and product movement. These newly created features helped in improving analytical insights and model performance.

5.2.2 Revenue Calculation:

Revenue was calculated by multiplying **units_sold** with **price_unit** for each observation. This derived variable represents the total sales value generated per transaction and is used for further analysis of financial performance.

5.2.3 Column Renaming:

To improve clarity and interpretation, the column **sku** was renamed to **Unique_ID**, representing a unique product identifier (e.g., MI-006 for milk products). This helps in better understanding and tracking of individual products throughout the analysis.

5.3 Descriptive Statistics

5.3.1 Descriptive Analysis

Variable	Count	Mean	Min	25%	50%	75%	Max	Std Dev
date	190754	28-10-2023	21-01-2022	16-04-2023	12-11-2023	07-06-2024	31-12-2024	-
price_unit	190754	5.25	1.5	3.38	5.25	7.13	9	2.17
promotion_flag	190754	0.15	0	0	0	0	1	0.36
delivery_days	190754	3	1	2	3	4	5	1.41
stock_available	190754	157.7	0	124	155	192	405	52.73
delivered_qty	190754	179.34	1	152	179	206	366	40.03
units_sold	190754	19.92	0	12	18	25	139	11.77
Year	190754	2023.29	2022	2023	2023	2024	2024	0.71
Month	190754	6.95	1	4	7	10	12	3.37
Day	190754	15.86	1	8	16	23	31	8.8
Revenue	190754	104.59	0	50.5	85.44	136.16	1145.6	79.52

Table 5.1: Descriptive Statistics

Descriptive Statistics Interpretation:

The dataset consists of **190,754 observations**, covering the period from **January 2022 to December 2024**, indicating a well-distributed time span for analysis. The average **price per unit is 5.25**, with values ranging from 1.5 to 9, showing moderate price variation across products.

On average, **units sold per transaction is around 19.92**, with a wide range (0 to 139), indicating variability in demand. The **average revenue is 104.59**, with high dispersion, suggesting differences in sales performance across products and time. Additionally, the **promotion flag mean of 0.15** shows that promotions are applied in about **15% of the observations**.

Inventory-related variables indicate that the average **stock available is 157 units**, while **delivered quantity averages 179 units**, reflecting consistent supply levels. Overall, the dataset shows moderate variability across key variables, making it suitable for detailed analysis and modeling.

5.3.2 Descriptive Analysis

- Price Unit Variance = 4.69

This indicates low variability in product prices, meaning prices are relatively stable across observations.

- Delivery Days Variance = 2.00

This shows very low variability, suggesting delivery time is consistent and well-controlled.

- Stock Available Variance = 2780.72

This indicates high variability in inventory levels, meaning stock fluctuates significantly across time and products.

- Delivered Quantity Variance = 1602.48

This reflects considerable variation in supply quantities, indicating inconsistency in deliveries.

- Units Sold Variance = 138.52

This shows moderate variability, suggesting that demand changes but remains within a reasonable range.

- Revenue Variance = 6323.81

This indicates very high variability in revenue, showing large fluctuations in sales performance.

Overall, the variability analysis indicates that **revenue and inventory-related variables exhibit high fluctuations**, suggesting dynamic changes in sales performance and stock levels across the dataset. In contrast, **price and delivery time show low variability**, indicating stable pricing strategies and consistent logistics operations.

This implies that while operational factors like pricing and delivery are well-controlled, **sales outcomes are more influenced by changing demand and inventory conditions**, making inventory management and demand variability key areas of focus in the analysis.

5.3.3 Mode Analysis (Categorical Insight)

- Mode of Region = PL-North
This indicates that most transactions are concentrated in the North region of Poland.
- Mode of Brand = SnBrand2
This shows that SnBrand2 is the most frequently occurring brand in the dataset.
- Mode of Segment = Yogurt-Seg1
This suggests that Yogurt-Seg1 is the most common product segment.
- Mode of Category = Yogurt
This indicates that yogurt products dominate the dataset in terms of frequency.
- Mode of Channel = Retail
This shows that the retail channel has the highest number of transactions.
- Mode of Year = 2024
This suggests that most data observations are from the year 2024.
- Mode of Pack Type = Carton
This indicates that carton packaging is the most frequently used format.
- Mode of Month = 12
This shows that December has the highest number of transactions.
- Mode of Day = 28
This indicates that the 28th day appears most frequently in the dataset.

Conclusive Insight

Overall, the dataset is dominated by **Yogurt category products**, particularly in the **Yogurt-Seg1 segment**, with most transactions occurring in the **Retail channel** and **PL-North region**. The data is also more concentrated in **2024 and December**, suggesting higher activity in recent periods and towards the year-end.

5.3.4 Distribution Analysis

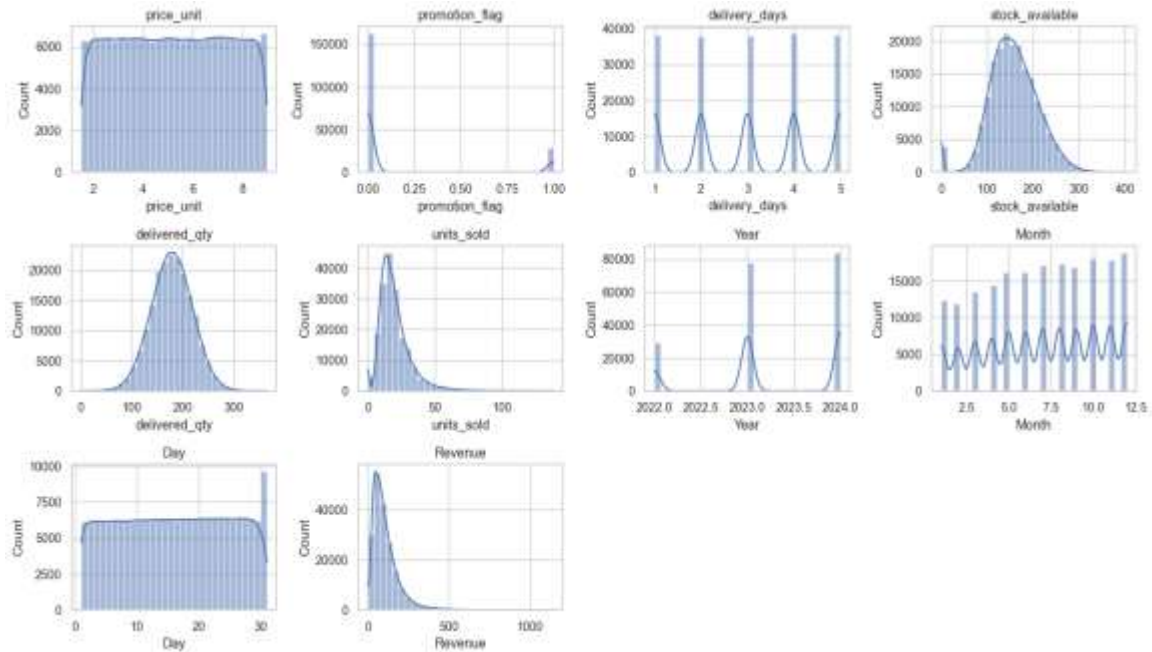


Figure 5.1: Histogram with KDE

The distribution of numerical variables was analyzed using histograms and density plots to understand data patterns and skewness.

- **Price Unit** shows a fairly uniform distribution, indicating prices are evenly spread across the range.
- **Units Sold** and **Revenue** are positively skewed, meaning most values are low to moderate with a few high-value observations.
- **Stock Available** and **Delivered Quantity** follow an approximately normal distribution, suggesting stable inventory patterns.
- **Promotion Flag** is highly imbalanced, with most observations being non-promotional (0).
- **Delivery Days** shows a discrete distribution, concentrated between 1 to 5 days.
- **Time variables (Year, Month, Day)** indicate consistent data coverage, with slight increases in recent periods.

Overall Interpretation:

The presence of skewness and non-normal distributions justifies the use of non-parametric methods and count-based models like the Negative Binomial model for further analysis.

5.3.5 Outlier Detection

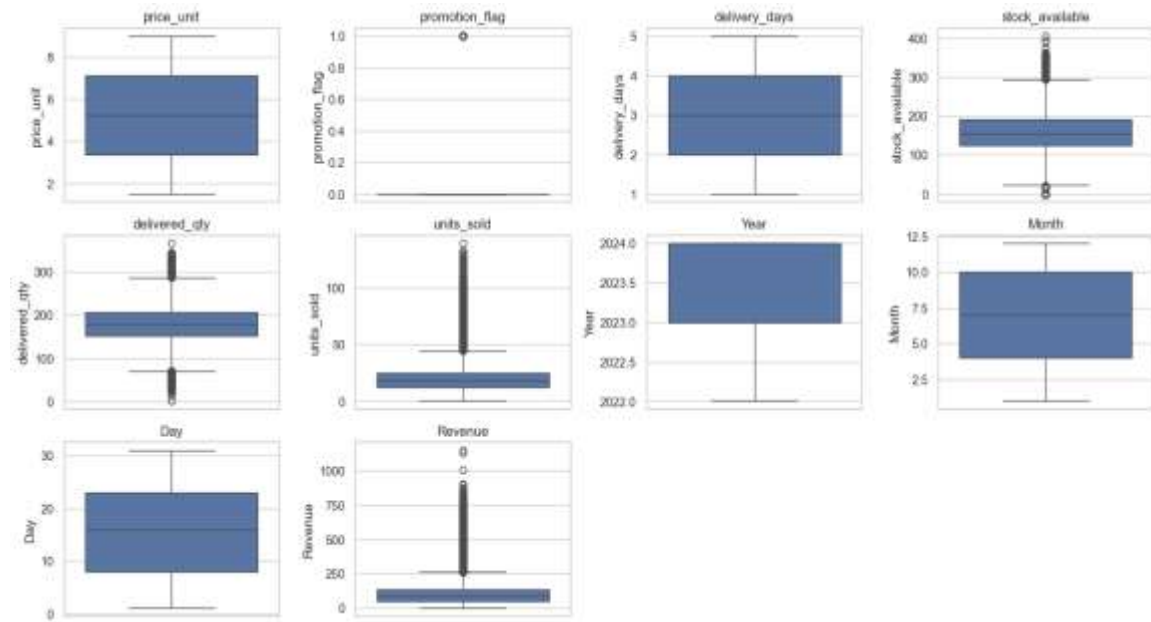


Figure 5.2: Box Plot Visualization of Dataset Features

Outliers in the numerical variables were identified using boxplots to examine the spread and extreme values in the dataset.

- **Units Sold** and **Revenue** show a significant number of upper-end outliers, indicating occasional high sales transactions.
- **Stock Available** and **Delivered Quantity** also contain some outliers, suggesting irregular inventory levels in certain cases.
- **Price Unit** and **Delivery Days** show relatively fewer outliers and are more consistently distributed.
- **Promotion Flag**, being a binary variable, does not exhibit meaningful outliers.

Overall Interpretation:

The presence of outliers reflects real-world variability in sales and inventory data. Since these values are meaningful and not data errors, they were retained for analysis.

6. OBJECTIVE 1: SALES PERFORMANCE ANALYSIS

6.1 Introduction

This objective focuses on analyzing the overall sales performance across different dimensions such as time, region, channel, category, brand, and pack type. The purpose is to understand how sales vary over time and across different segments.

Various techniques including trend analysis, distribution analysis, contribution analysis, and correlation analysis are used to identify key patterns and insights. This helps in understanding the overall sales behaviour, identifying top-performing segments, and evaluating factors influencing sales performance.

6.2 Time-Based Sales Analysis

6.2.1 Year-wise Sales Trend

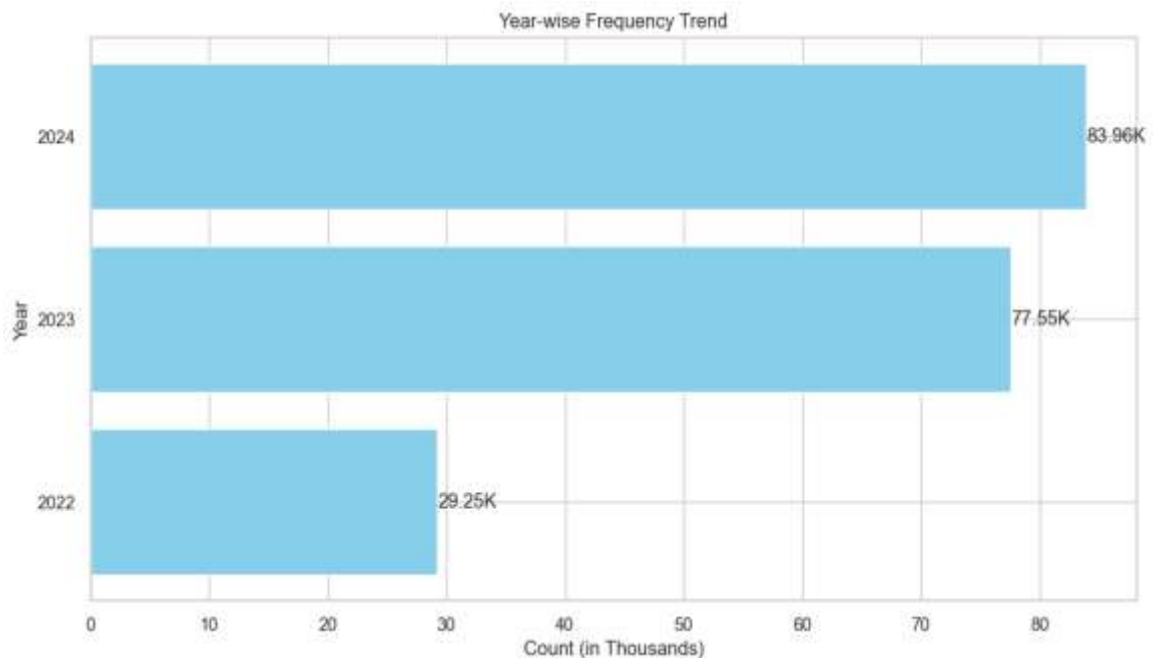


Figure 6.1: Year-wise Sales Trend

The horizontal bar chart illustrates the distribution of sales records across years. It is observed that 2024 has the highest count (~83.96K), followed by 2023 (~77.55K), while 2022 has the lowest (~29.25K).

The visual clearly shows a **progressive increase from 2022 to 2024**, indicating a rise in sales activity or data coverage over time. The shorter bar for 2022 suggests relatively limited observations, whereas the longer bars for 2023 and 2024 reflect higher sales presence.

Interpretation:

The graph highlights a consistent upward trend, suggesting that sales performance or data volume has increased in recent years, with 2024 being the most significant contributor.

6.2.2 Month-wise Sales Trend

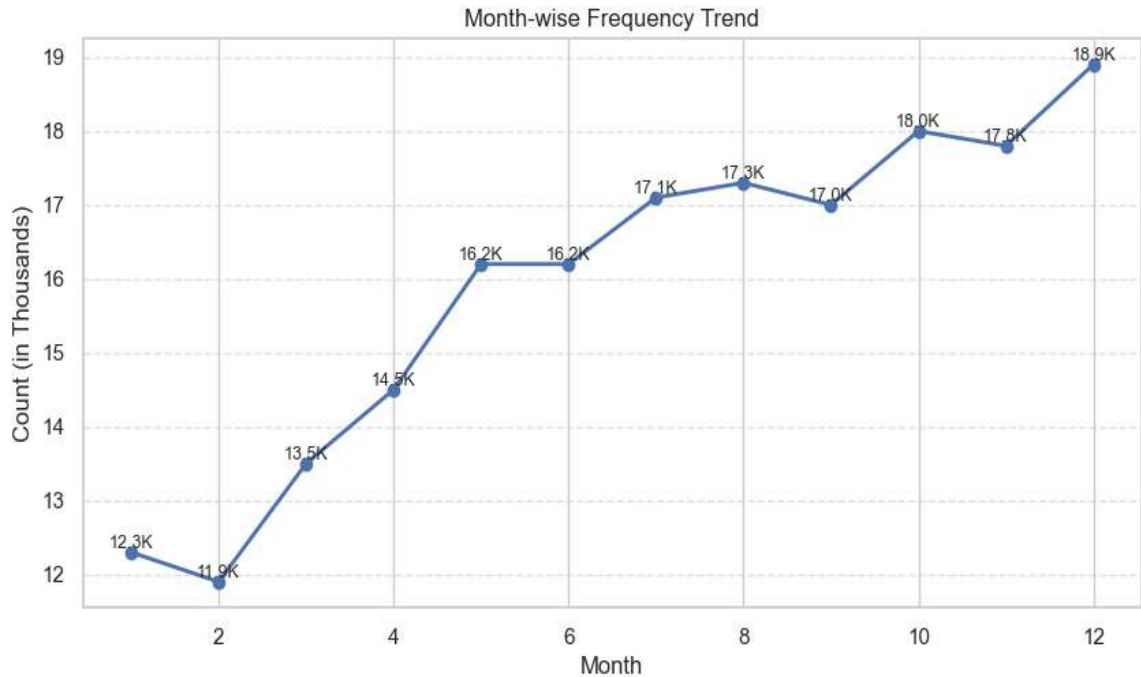


Figure 6.2: Month-wise Sales Trend

The line chart shows the distribution of sales records across months. It is observed that sales gradually increase from the beginning of the year, starting at **12.3K in January**, with a slight dip in **February (11.9K)**, and then steadily rising throughout the year.

Sales activity peaks in the later months, with **December recording the highest value (~18.9K)**, followed by **October (18.0K)** and **November (17.8K)**. The upward slope of the line indicates a consistent increase in sales frequency as the year progresses.

Interpretation:

The graph suggests a seasonal pattern where sales are relatively lower in the early months and higher toward the end of the year, indicating increased demand or activity in later months.

6.2.3 Day-wise Sales Pattern

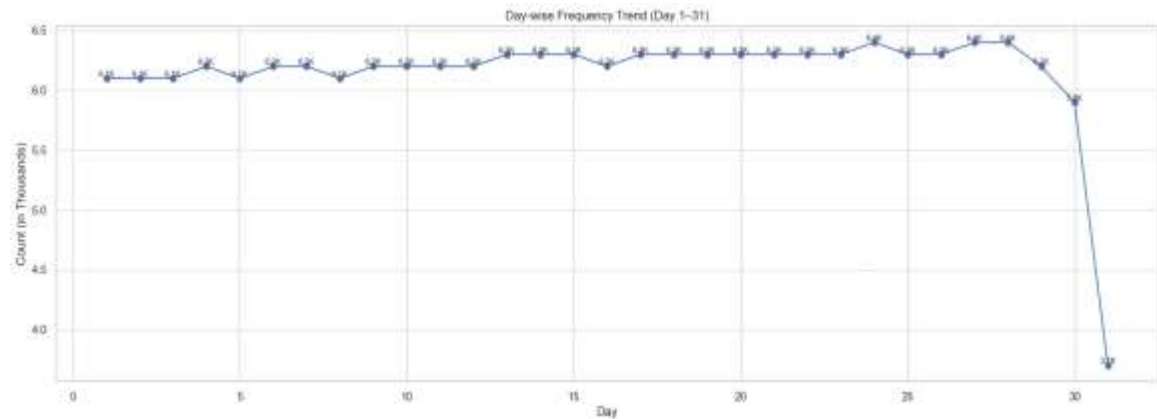


Figure 6.3: Day-wise Sales Pattern

The day-wise line graph shows that sales frequency remains relatively stable throughout most days of the month, generally ranging between **6.1K to 6.4K**. There is a slight increase in the middle and later days, with peak values observed around **Day 24, 27, and 28 (~6.4K)**. However, a noticeable decline is seen towards the end of the month, particularly on **Day 30 (5.9K)** and **Day 31 (3.7K)**, which records the lowest frequency.

Interpretation:

The pattern indicates consistent daily sales activity across the month with minor fluctuations. The sharp drop on the 31st is expected due to fewer occurrences of this day in the calendar, rather than an actual decline in demand.

6.3 Sales Distribution Across Dimensions



Figure 6.4: Region-wise Sales Distribution Figure 6.5: Channel-wise Sales Distribution

6.3.1 Region-wise Sales Distribution

The donut chart shows that sales are almost evenly distributed across all regions. **PL-North (1.27M)** records the highest sales, followed closely by **PL-South (1.268M)** and **PL-Central (1.262M)**.

Interpretation:

There is no significant regional variation in sales, indicating a balanced market presence across all regions.

6.3.2 Channel-wise Sales Distribution

The donut chart indicates that sales are nearly evenly distributed across all channels. **E-commerce (1.269M)** has the highest sales, followed closely by **Retail (1.267M)** and **Discount (1.264M)**.

Interpretation:

There is no major difference in sales performance across channels, suggesting that all channels contribute almost equally to overall sales.

6.3.3 Pack Type Analysis



Figure 6.6: Pack Type Analysis

The pie chart shows that sales are almost equally distributed across pack types. **Single packs (1.27M)** have the highest sales, followed by **Carton (1.267M)** and **Multipack (1.263M)**.

Interpretation:

There is minimal variation in sales across pack types, indicating similar consumer preference for all packaging formats.

6.4 Product-Level Analysis

6.4.1 Category-wise Sales

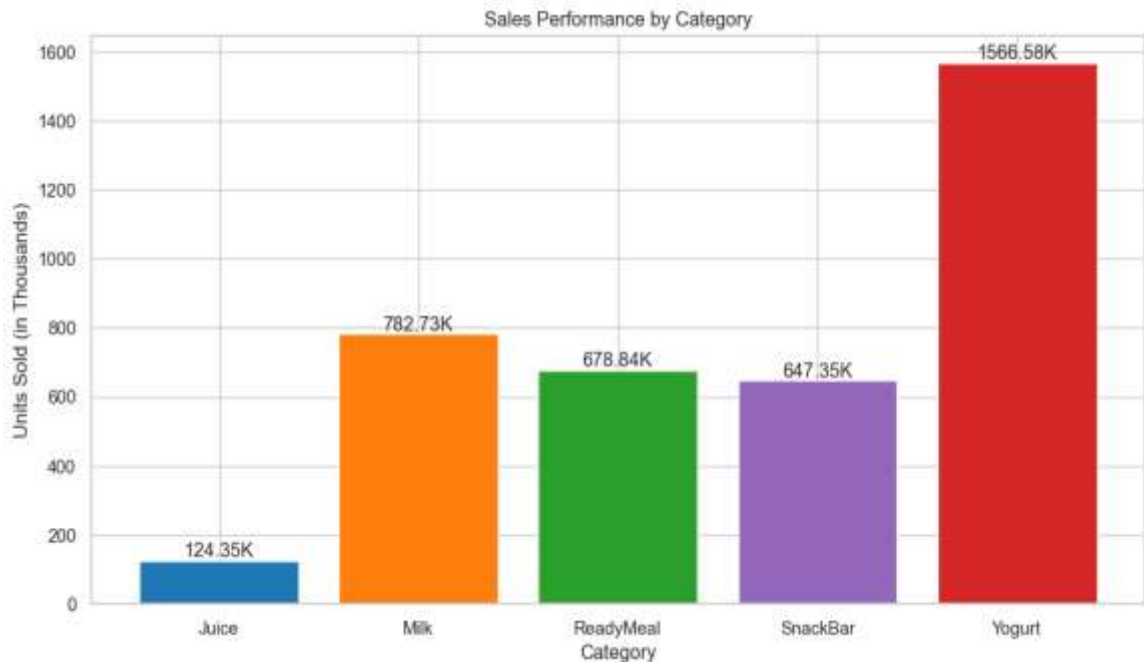


Figure 6.7: Category-wise Sales

The bar chart indicates significant variation in sales across product categories. **Yogurt (1566.6K)** clearly dominates the market, contributing nearly double the sales compared to the next category, **Milk (782.7K)**. Other categories such as **ReadyMeal (678.8K)** and **SnackBar (647.4K)** show moderate and relatively similar performance. In contrast, **Juice (124.3K)** records the lowest sales among all categories.

Interpretation:

The results suggest that Yogurt is the most in-demand category, playing a major role in overall sales performance. Milk, ReadyMeal, and SnackBar contribute steadily but at a lower level, while Juice shows comparatively weak demand, indicating lower consumer preference within the dataset.

6.4.2 Brand-wise Sales

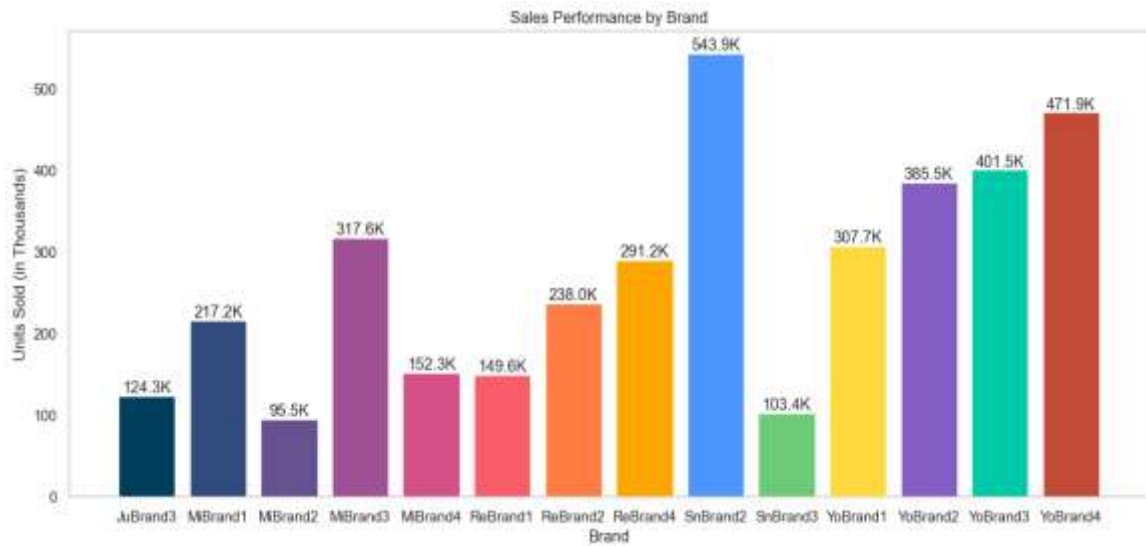


Figure 6.8: Brand-wise Sales

The bar chart shows variation in sales performance across brands. **SnBrand2 (543.9K)** records the highest sales, followed by **YoBrand4 (471.9K)** and **YoBrand3 (401.5K)**. Several brands such as **YoBrand2 (385.5K)**, **MiBrand3 (317.6K)**, and **YoBrand1 (307.7K)** also contribute significantly to overall sales.

On the lower end, brands like **JuBrand3 (124.3K)**, **SnBrand3 (103.4K)**, and **MiBrand2 (95.5K)** show comparatively lower performance.

Interpretation:

Sales are moderately concentrated among top-performing brands, with a few brands contributing a larger share, while others have relatively lower demand. This indicates variation in brand popularity and market performance within the dataset.

6.5 Contribution Analysis

6.5.1 Category Contribution (%)

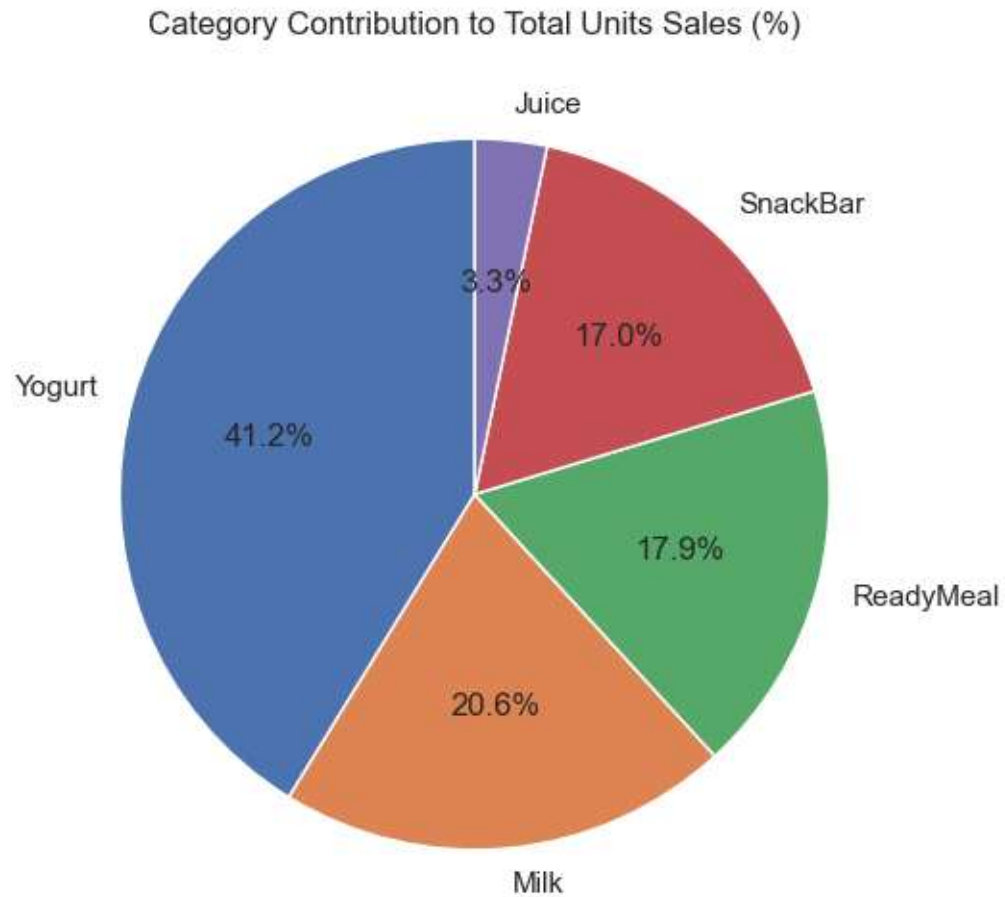


Figure 6.9: Category Contribution (%)

The pie chart highlights the percentage contribution of each category to total sales. **Yogurt (41.23%)** holds the largest share, followed by **Milk (20.6%)**, **ReadyMeal (17.86%)**, and **SnackBar (17.04%)**. **Juice (3.27%)** contributes the least.

Interpretation:

Sales are heavily dominated by the Yogurt category, contributing over 40% of total sales. The remaining categories share a moderate portion, while Juice has a minimal impact on overall sales.

6.5.2 Brand Contribution (%)

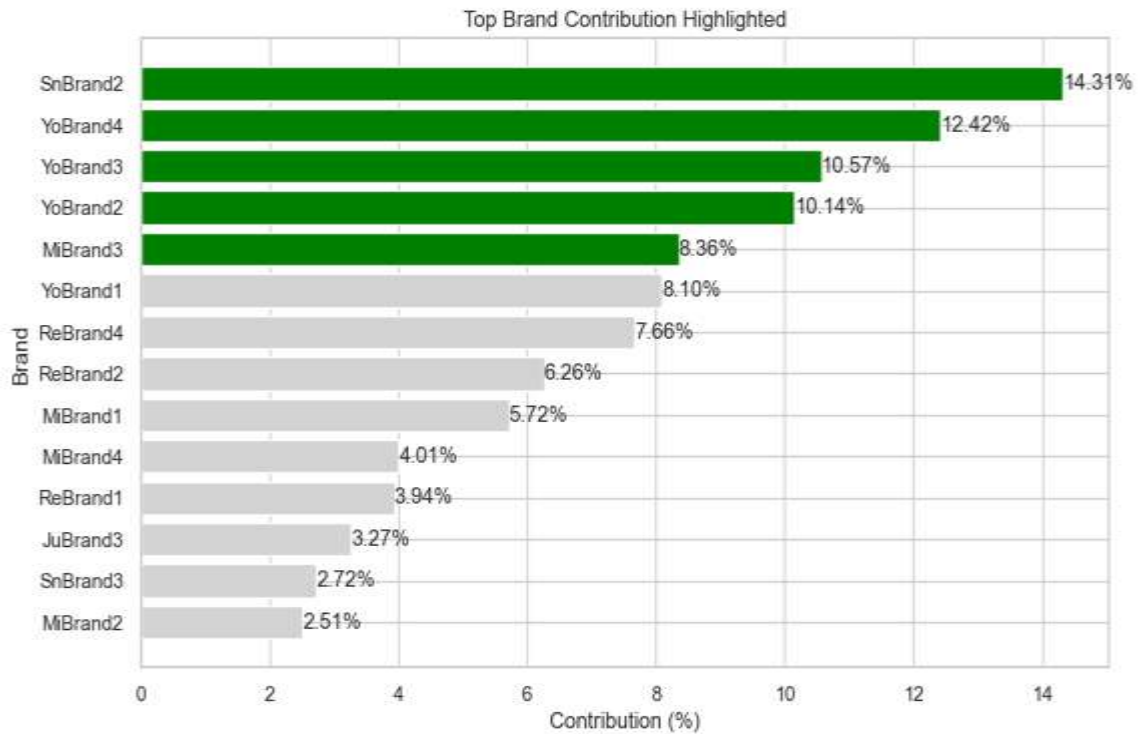


Figure 6.10: Brand Contribution (%)

The horizontal bar chart highlights the contribution of brands to total sales, with the top 5 brands clearly standing out. **SnBrand2 (14.31%)** contributes the highest share, followed by **YoBrand4 (12.42%)**, **YoBrand3 (10.57%)**, **YoBrand2 (10.14%)**, and **MiBrand3 (8.36%)**.

The combined contribution of these top 5 brands is approximately **55.80%**, indicating that more than half of the total sales are generated by a small group of leading brands.

Interpretation:

Sales are concentrated among the top-performing brands, with the top 5 contributing a majority share. This suggests strong brand dominance, while the remaining brands contribute relatively smaller portions to overall sales.

6.6 Correlation Analysis

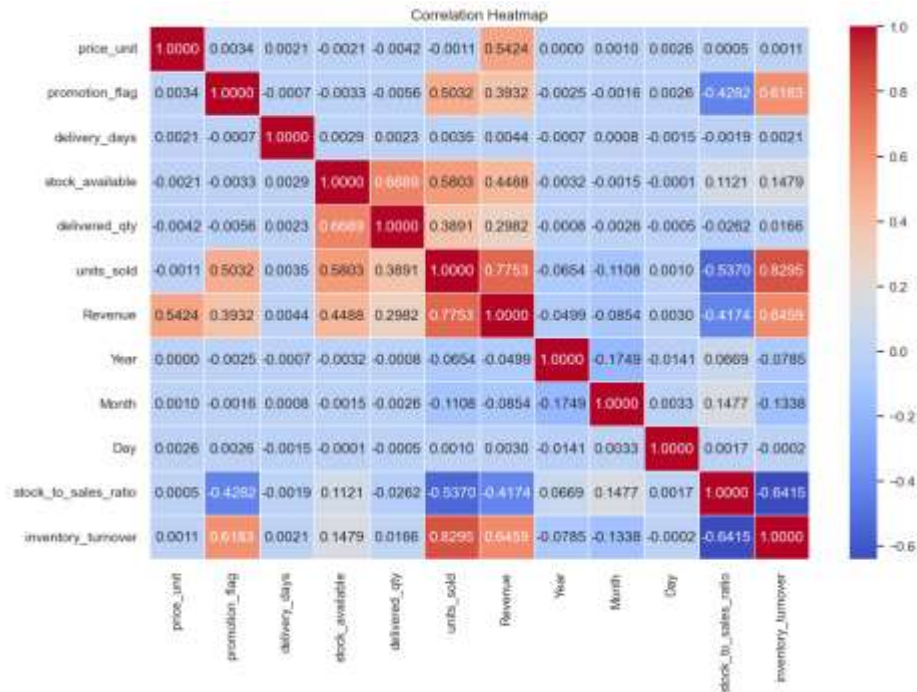


Figure 6.11: Heatmap

- **Units_sold** and **Revenue** show a **strong positive correlation** (~0.77), indicating that higher sales directly increase revenue.
- **Inventory_turnover** is highly positively correlated with **units_sold** (~0.83), suggesting efficient stock usage leads to higher sales.
- **Stock_available** has a moderate positive relationship with **units_sold** (~0.58), meaning higher stock supports sales.
- **Promotion_flag** shows a moderate positive correlation with **units_sold** (~0.50), indicating promotions help increase sales.
- **Stock_to_sales_ratio** has a **negative correlation** with **units_sold** (~-0.54), suggesting excess stock relative to sales may indicate inefficiency.
- Most other variables show **weak or negligible correlations**, indicating limited direct relationships.

Overall Interpretation:

Sales performance is mainly influenced by **inventory efficiency, stock levels, and promotions**, while other variables have minimal direct impact.

6.7 Heatmap Analysis

6.7.1 Monthly Units Sold by Category

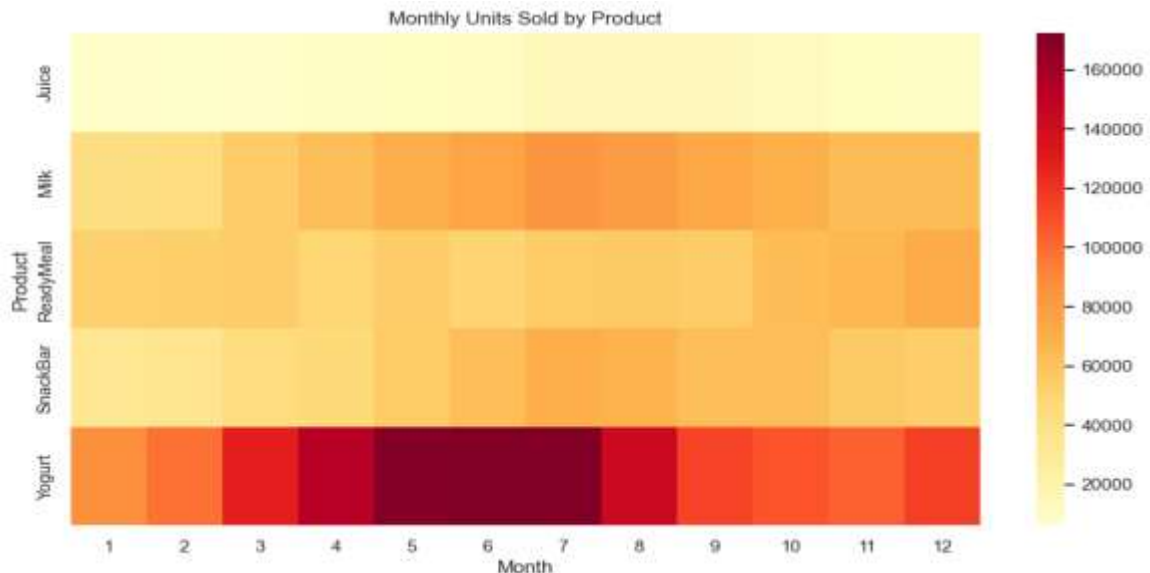


Figure 6.12: Monthly Units Sold by Category

The heatmap presents the distribution of units sold across different product categories over the months.

- **Yogurt** shows the highest sales intensity across almost all months, with peak performance observed between **April and July**, indicating strong demand during mid-year.
- **Milk** demonstrates a relatively stable trend with a slight increase in sales during the middle months, followed by a gradual decline towards the end of the year.
- **ReadyMeal** and **SnackBar** categories exhibit moderate sales with mild seasonal variation, showing slight improvements in the later months.
- **Juice** consistently records the lowest sales among all categories, with minimal fluctuation throughout the year.

Overall Interpretation:

The heatmap clearly highlights **seasonality in sales**, where most categories experience higher demand during mid-year months. Yogurt dominates the market, while Juice remains a low-performing category. This insight can help in planning inventory, promotions, and demand forecasting strategies.

6.8 Time Series Analysis

6.8.1 Overall Sales Trend

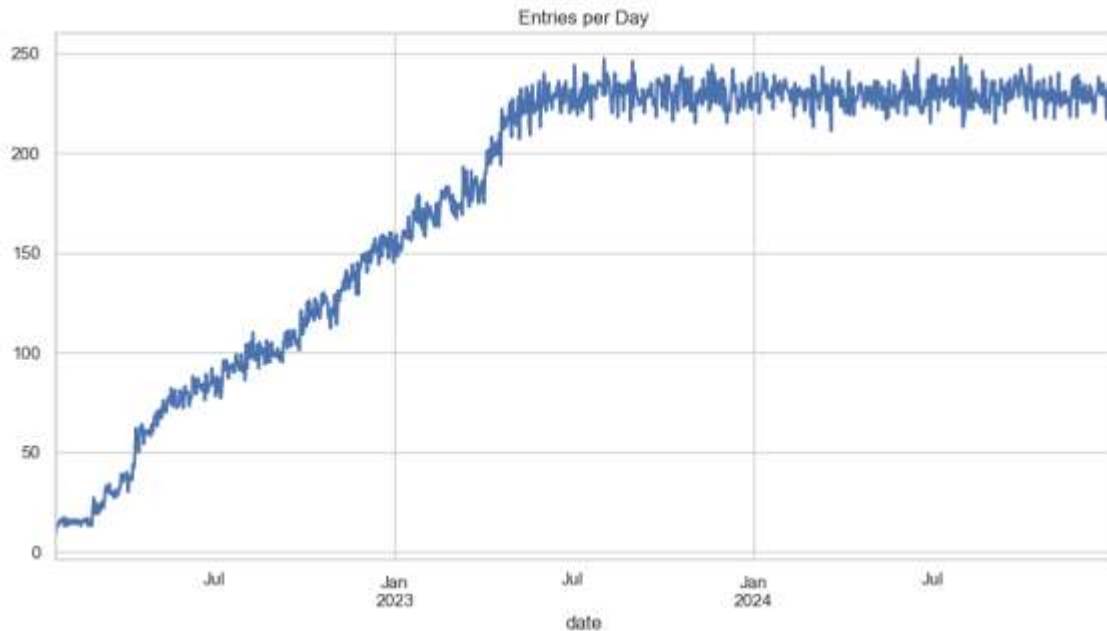


Figure 6.13: Entries Per Day

Entries Per Day:

- The dataset contains **190,754 records** spanning from **January 2022 to December 2024**, providing a continuous daily time series.
- The “Entries per Day” plot shows a **clear upward trend** from early 2022 to mid-2023, indicating a gradual increase in sales activity or data volume over time.
- After mid-2023, the trend becomes **stable**, with daily entries fluctuating around a consistent level (approximately 220–240), suggesting a mature and steady sales pattern.
- Small ups and downs are seen in the data, which are normal daily changes and not large or unusual variations.

Overall Interpretation:

The time series exhibits an initial **growth phase followed by stabilization**, indicating that sales or recorded transactions increased over time and then reached a steady state. This pattern reflects realistic business behavior, where growth is followed by consistency, making the dataset suitable for trend and forecasting analysis.

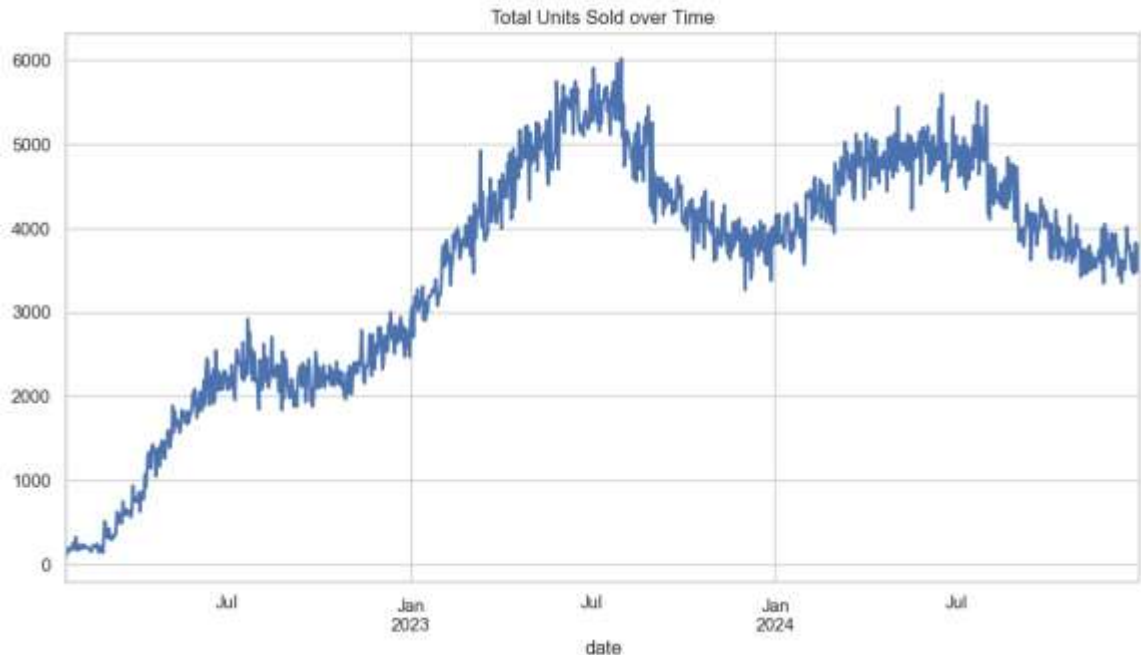


Figure 6.14: Total Units Sold Over Time

6.8.2 Total Units Sold Over Time:

- The plot shows the **daily total units sold** across the entire time period.
- There is a **gradual increasing trend** in sales from 2022 to mid-2023, indicating growing demand over time.
- After this period, sales become more **stable**, with consistent fluctuations around a higher level.
- Regular ups and downs are visible, which represent **normal daily changes in sales**.

Overall Interpretation:

The trend indicates that sales **increased over time and then stabilized**, suggesting business growth followed by a steady demand phase. This pattern is useful for understanding long-term performance and planning future sales strategies.

6.8.3 Category-wise Trend

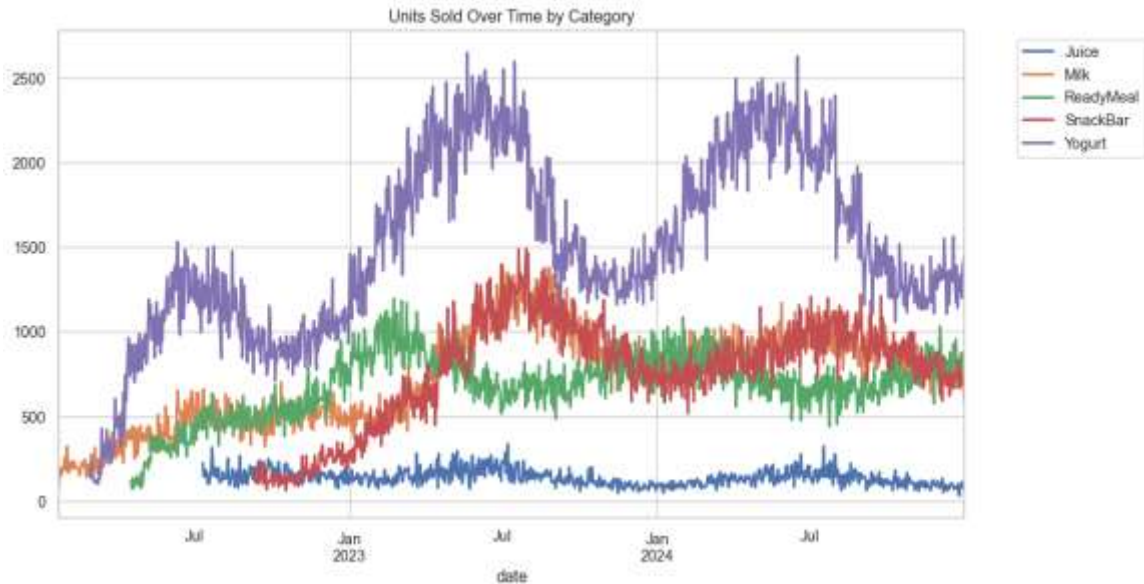


Figure 6.15: Category-wise Trend

The line chart shows the monthly units sold for five product categories (Juice, Milk, ReadyMeal, SnackBar, and Yogurt) from mid-2022 to mid-2024.

- Yogurt is the highest-selling category overall, with strong peaks reaching over 2,500 units, especially during mid-2023 and early 2024. Sales remain relatively high but show a gradual decline after mid-2024.
- SnackBar and Milk follow as the next strongest performers, both showing steady growth until mid-2023, with sales stabilizing between 800–1,200 units.
- ReadyMeal shows moderate performance with visible fluctuations, maintaining sales mostly between 600–1,000 units.
- Juice has the lowest sales volume, consistently staying below 400 units throughout the period, with very little growth.

Overall Trend:

Yogurt dominates sales but has started showing signs of slowdown. SnackBar and Milk appear more stable. Juice remains a low-performing category.

6.9 Conclusion for Objective 1: Sales Performance Analysis

The analysis shows that sales **increased from 2022 to mid-2023 and then became stable**, indicating consistent demand. A clear **seasonal pattern** is observed, with higher sales in mid-year months, while daily sales remain mostly stable.

Yogurt is the top-performing category, and a few brands like **SnBrand2 and YoBrand series** contribute a large share of total sales. Sales are **evenly distributed across regions, channels, and pack types**, showing a balanced market.

The data also shows some **skewness and outliers**, meaning occasional high sales values are present. Correlation results indicate that **units sold strongly drives revenue**, and factors like **stock availability, inventory turnover, and promotions positively impact sales**.

Overall, sales performance is stable and mainly driven by **product demand, efficient inventory, and promotions**.

7. OBJECTIVE 2: PRICE IMPACT ON SALES

7.1 Introduction

This objective focuses on analyzing the relationship between price per unit and sales performance (units sold and revenue). The aim is to evaluate whether changes in price influence customer demand and overall sales outcomes.

Various statistical techniques and visualizations, such as correlation analysis, scatter plots, price group comparison, and non-parametric tests, are used to examine the impact of price on sales. This helps in understanding whether pricing plays a significant role in driving sales or if other factors have a greater influence.

7.2 Correlation Analysis (Pearson)

The Pearson correlation between **price per unit and units sold** is **-0.001**, which is very close to zero.

This indicates that there is **no meaningful linear relationship** between price and sales. Changes in price do not significantly affect the number of units sold, and the relationship between the two variables is negligible.

7.3 Scatter Plot Analysis

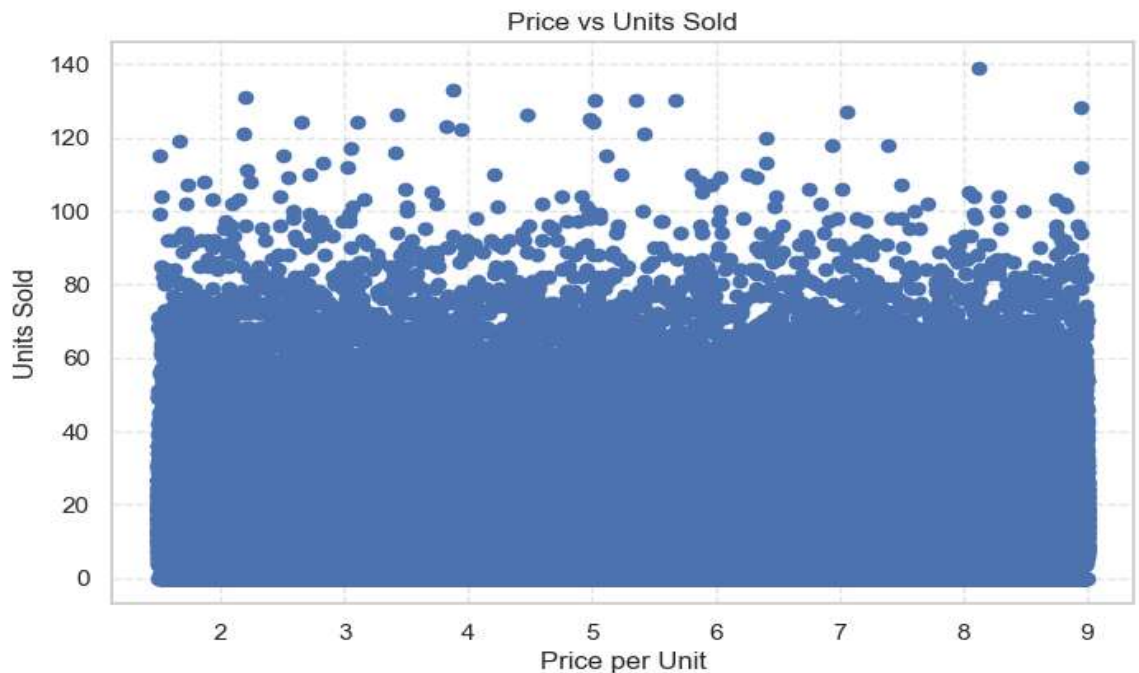


Figure 7.1: Scatter Plot of Price Vs Units Sold

The scatter plot shows a **random distribution of data points** across different price levels. There is no clear upward or downward pattern between price and units sold.

Sales remain **spread across all price ranges**, indicating that both low and high-priced products have similar sales levels. The absence of a clear trend suggests that **price does not strongly influence sales behaviour** in the dataset.

7.4 Spearman Correlation

Null Hypothesis (H_0): There is no monotonic relationship between price_unit and units_sold.

Alternative Hypothesis (H_1): There is a significant monotonic relationship between price_unit and units_sold.

- Spearman Correlation Coefficient: -0.0008915315272088063
- P-value: 0.6969969664572935

Interpretation:

This indicates a **very weak and statistically insignificant relationship** between price and sales. The high p-value (> 0.05) suggests that the relationship is not statistically significant.

7.5 Price Group Analysis

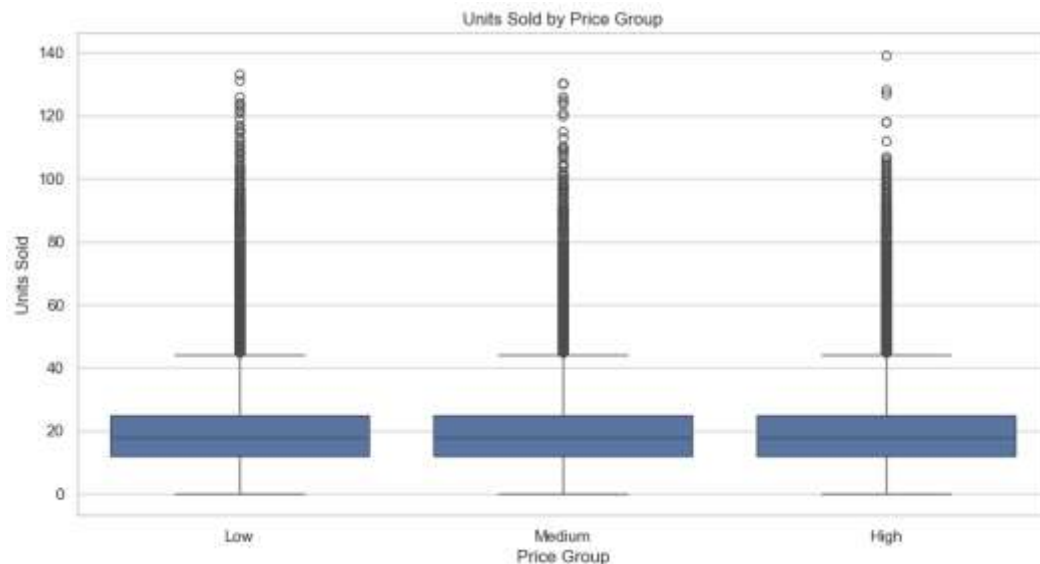


Figure 7.2: Box Plot of Units Sold by Price Group

The boxplot shows that **units sold are quite similar across Low, Medium, and High price groups**, as all groups have nearly the same median values. The spread of data (range and IQR) is also comparable, indicating consistent sales distribution across different price levels.

Although some outliers (high sales values) are present in all groups, they are evenly distributed and do not favour any particular price category. Overall, this suggests that **price range does not have a strong impact on units sold**, and sales remain relatively stable regardless of price differences.



Figure 7.3: Average Sales by Price Range

The bar chart shows that the **average units sold are almost identical across Low, Medium, and High price ranges**, all around the same level (~20 units). There is no noticeable difference in average sales between the price groups.

This indicates that **price range does not significantly influence average sales**, as all categories perform at a similar level.

7.6 Kruskal-Wallis Test

7.6.1 Hypothesis

Null Hypothesis (H_0): There is no significant difference in the distribution (median) of the variable across the Low, Medium, and High groups.

Alternative Hypothesis (H_1): There is a significant difference in the distribution (median) among at least one of the groups (Low, Medium, High).

- **Test Statistic:** 0.348
- **P-value:** 0.840

Interpretation:

The p-value is much greater than 0.05, indicating that there is **no statistically significant difference** in units sold across the three price groups. This means that sales remain similar regardless of the price range.

7.7 Conclusion for Objective 2: Price Impact on Sales

The analysis shows that there is **no significant relationship between price and units sold**. Both Pearson and Spearman correlation results indicate a **near-zero relationship**, and the scatter plot confirms the absence of any clear pattern. Additionally, the boxplot and average sales analysis show that **units sold are similar across all price groups**.

The Kruskal-Wallis test also indicates **no statistically significant difference** in sales between low, medium, and high price categories.

It is important to note that the dataset is **simulated**, which explains the lack of relationship between price and sales. Therefore, **further modelling or advanced statistical analysis based on price is not meaningful**, as price does not have a measurable impact on sales in this dataset.

8. OBJECTIVE 3: EFFECT OF PROMOTIONS ON SALES

8.1 Introduction

Promotions play a crucial role in influencing consumer purchasing behaviour in the FMCG sector. Businesses frequently use promotional strategies such as discounts, offers, and campaigns to boost sales and increase market share. However, it is important to evaluate whether these promotions actually lead to a significant increase in sales and revenue.

The objective of this analysis is to examine the impact of promotional activities on product performance, specifically in terms of units sold and revenue. This study compares promoted and non-promoted products using descriptive analysis, visualization techniques, and statistical tests to determine whether promotions significantly affect sales outcomes.

In addition, a Negative Binomial regression model is employed to quantify the effect of promotions while controlling for other factors such as stock availability. This helps in understanding not only whether promotions influence sales, but also the extent of their impact.

8.2 Descriptive Comparison of Promoted vs Non-Promoted Products

Promotion Flag	Avg Units Sold	Avg Revenue
0 (No Promotion)	17.44	91.5
1 (Promotion)	34.06	179.26

Table 8.1: Descriptive Comparison of Promoted vs Non-Promoted Products

The comparison shows that when **no promotion is applied (0)**, the average units sold are **17.44** with an average revenue of **91.50**. In contrast, when **promotion is applied (1)**, the average units sold increase to **34.06**, and average revenue rises to **179.26**.

Promoted products have **significantly higher sales and revenue** compared to non-promoted products. This indicates that promotions are effective in boosting both **demand and overall sales performance**.

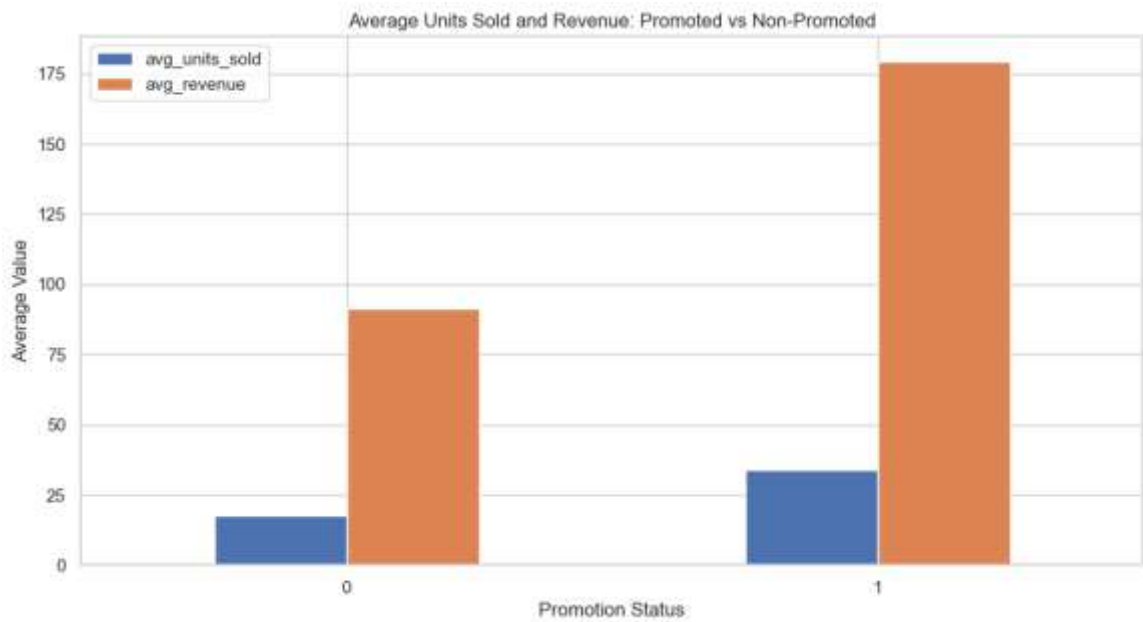


Figure 8.1: Average Units Sold & Revenue – Promoted Vs Non Promoted

The bar chart clearly shows that **both average units sold and average revenue are significantly higher for promoted products (1)** compared to non-promoted products (0). Promoted products have **almost double the sales and revenue**, highlighting a strong difference between the two groups.

The visualization confirms that **promotions have a strong positive impact on both sales and revenue**, making them an effective strategy to increase product performance.

8.3 Distribution Comparison Using Boxplots

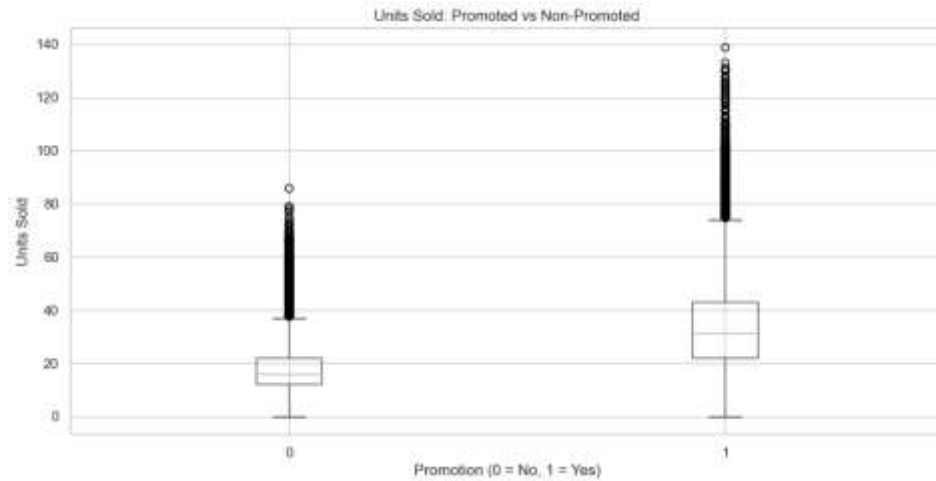


Figure 8.2: Distribution Comparison Using Boxplots

The boxplot shows that **promoted products (1) have a higher median and wider range of units sold** compared to non-promoted products (0). The entire distribution for promoted products is shifted upward.

Additionally, promoted products show **more high-value outliers**, indicating instances of very high sales during promotions.

Promotions not only increase average sales but also lead to **higher and more variable sales performance**, with a greater chance of achieving high sales values.

8.4 Mann–Whitney U Test

8.4.1 Hypothesis

Null Hypothesis (H_0): There is no significant difference in the distribution of units_sold between promoted and non-promoted products.

Alternative Hypothesis (H_1): There is a significant difference in the distribution of units_sold between promoted and non-promoted products.

- **Test Statistic:** 3834404037.5
- **P-value:** 0.0

Interpretation:

The p-value is less than 0.05, indicating a **statistically significant difference** in units sold between promoted and non-promoted products. This means that the distribution of sales for promoted products is **significantly higher** than for non-promoted products.

8.5 Promotion Impact Across Regions

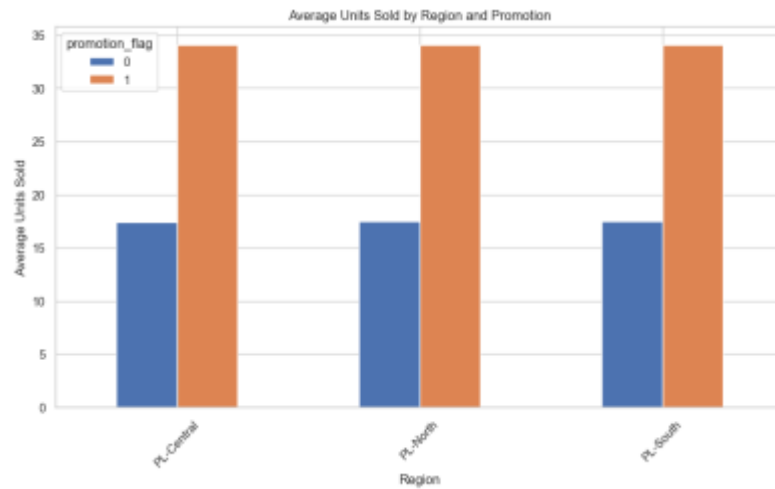


Figure 8.3: Promotion Impact Across Regions

The bar chart shows that in all regions (**PL-Central**, **PL-North**, **PL-South**), promoted products have higher average sales than non-promoted ones. The increase is consistent across all regions.

Promotions have a positive and similar impact on sales in every region.

8.6 Promotion Impact Across Channels

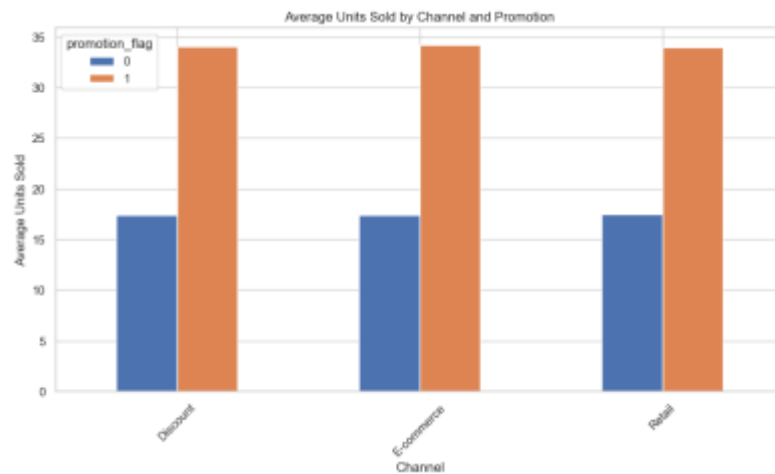


Figure 8.4: Promotion Impact Across Channels

The bar chart shows that in all channels (**Discount**, **E-commerce**, **Retail**), **promoted products have higher average sales** than non-promoted ones.

Promotions consistently increase sales across all channels, showing a similar positive effect everywhere.

8.7 Negative Binomial Regression Analysis

8.7.1 Justification for Model Selection

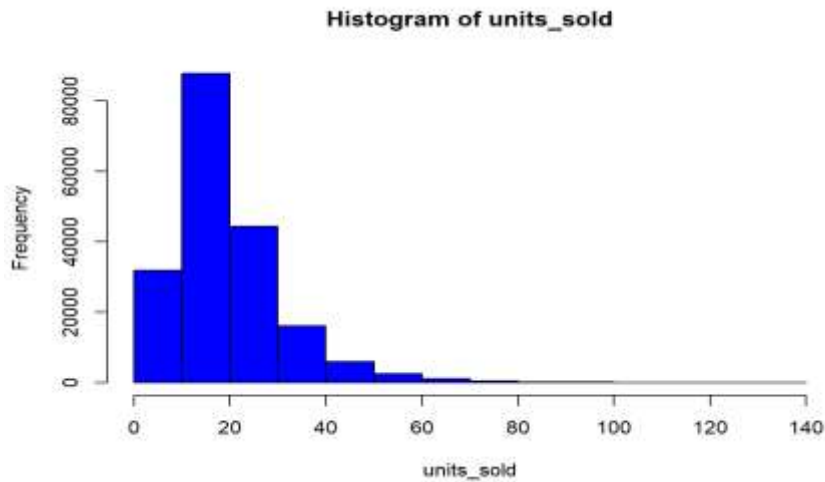


Figure 8.5: Histogram of Units Sold

8.7.2 Histogram of units_sold:

The dependent variable **units_sold** represents count data, as it takes non-negative integer values (minimum = 0, maximum = 139). The histogram shows a **right-skewed distribution**, indicating that most observations are concentrated at lower values with a long tail of higher values.

8.7.3 Mean and variance check:

To assess dispersion, the **mean (19.92)** and **variance (138.52)** were compared. Since the variance is significantly higher than the mean, this indicates the presence of **overdispersion** in the data.

8.7.4 Poisson dispersion result:

A Poisson regression model was initially fitted; however, the **dispersion statistic (~1.77)** is greater than 1, confirming that the Poisson model assumptions are violated due to overdispersion.

Interpretation:

Since the data exhibits overdispersion and skewness, the Poisson model is not suitable. Therefore, a **Negative Binomial regression model** is more appropriate, as it can handle overdispersed count data effectively.

8.7.2 Final Model Results

```
##
## Call:
## glm.nb(formula = units_sold ~ promotion_flag + stock_available,
##       data = df, init.theta = 15.25744996, link = log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.668e+00  2.732e-03   610.3  <2e-16 ***
## promotion_flag  6.735e-01  2.052e-03   328.1  <2e-16 ***
## stock_available 7.140e-03  1.548e-05   461.2  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(15.2574) family taken to be 1)
##
##      Null deviance: 529668  on 190753  degrees of freedom
## Residual deviance: 208710  on 190751  degrees of freedom
## AIC: 1243076
##
## Number of Fisher Scoring iterations: 1
##
##
##              Theta:  15.2574
##              Std. Err.:  0.0891
##
## 2 x log-likelihood: -1243068.4310
```

Figure 8.6: Final Model Results

A Negative Binomial regression model was fitted to examine the effect of **promotion_flag** and **stock_available** on **units_sold**.

The results show that **promotion_flag** has a **strong positive and statistically significant effect on sales** ($p < 0.001$). The coefficient value (**0.6735**) is interpreted using exponentiation:

$$e^{0.6735} \approx 1.96$$

This indicates that when a product is promoted, the expected units sold are **approximately 1.96 times higher**, meaning nearly a **96% increase in sales** compared to non-promoted products.

The variable **stock_available** (**0.00714**, $p < 0.001$) also shows a positive and significant effect, suggesting that higher stock availability leads to increased sales, although its impact is smaller compared to promotions.

8.7.3 Model Fit:

The model shows a strong fit, as the **residual deviance (208710)** is substantially lower than the **null deviance (529668)**, indicating that the model explains a significant portion of variation in sales. The **AIC value (1243076)** supports model adequacy, and the **theta value (15.26)** confirms that overdispersion is present and appropriately handled by the Negative Binomial model.

Overall Interpretation:

The results clearly demonstrate that **promotions have a significant and substantial impact on sales**, nearly doubling the expected units sold. While stock availability also contributes positively, **promotion is the dominant factor influencing sales performance**, making it a highly effective strategy for increasing demand.

8.7.4 Multicollinearity Check

The Variance Inflation Factor (VIF) values for the independent variables are:

- promotion_flag: 1.00067
- stock_available: 1.00067

Interpretation:

Since both VIF values are very close to 1 and well below the threshold of 5, this indicates that there is **no multicollinearity between the variables**. The predictors are independent of each other, and the model estimates are reliable.

8.8 Model Performance Evaluation (Weekly Model)

```
##
## Call:
## glm.nb(formula = units_sold ~ promotion_flag + stock_available +
##       lag_1, data = ts_weekly, init.theta = 14.49522483, link = log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.526e+00  1.698e+00   0.899   0.369
## promotion_flag -2.380e+00  1.743e+00  -1.365   0.172
## stock_available  4.688e-02  1.072e-02   4.375 1.22e-05 ***
## lag_1          5.812e-05  2.156e-06  26.960 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(14.4952) family taken to be 1)
##
##    Null deviance: 703.64  on 153  degrees of freedom
## Residual deviance: 156.09  on 150  degrees of freedom
## AIC: 3105.6
##
## Number of Fisher Scoring iterations: 1
##
##
##              Theta: 14.50
##            Std. Err.: 1.64
##
## 2 x log-likelihood: -3095.597
```

Figure 8.7: Model Performance Evaluation (Weekly Model)

A weekly Negative Binomial model was fitted by aggregating the data and including a **lag variable (lag_1)**, which represents the **previous week's sales**, to capture time dependency in sales.

8.8.1 Key Results:

- **Lag_1 ($p < 0.001$):**
The previous week's sales have a **strong and highly significant positive effect** on current sales. This indicates that sales follow a pattern where **past performance influences future demand**.
- **Stock_available ($p < 0.001$):**
Stock availability is also **positively significant**, showing that higher stock levels support higher sales even at the weekly level.

- Promotion_flag ($p = 0.172$):

Promotion is **not statistically significant** in this model, suggesting that its effect becomes less visible when data is aggregated weekly.

8.8.2 Model Fit:

- The **residual deviance (156.09)** is much lower than the **null deviance (703.64)**, indicating a strong improvement in model performance.
- The **AIC value (3105.6)** supports the suitability of the model.
- The **theta value (14.50)** confirms that overdispersion is present and properly handled.

Overall Interpretation:

The results show that **sales are strongly influenced by past sales (lag effect) and stock availability**, highlighting the importance of time dependency and inventory management. While promotions were significant at the daily level, their impact is reduced in the weekly model, indicating that **sales patterns are more driven by historical trends when viewed over time**.

8.8.2 Weekly Actual vs Predicted Sales plot

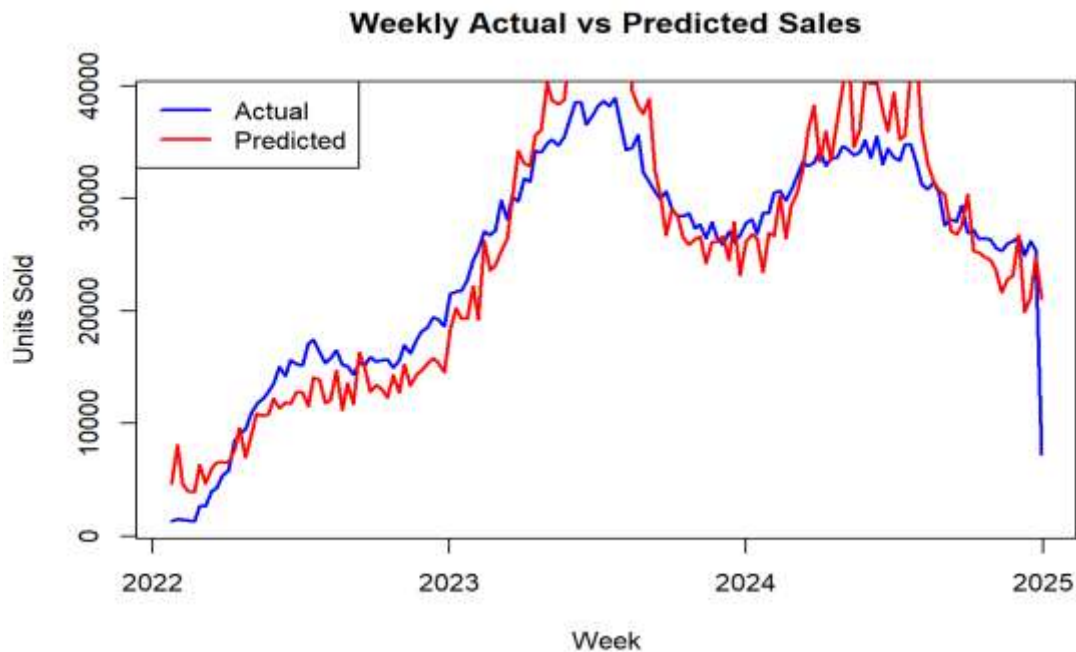


Figure 8.8: Weekly Actual vs Predicted Sales plot

The line plot compares the **actual sales (blue line)** with the **predicted sales (red line)** over time.

- The predicted values generally **follow the overall trend** of actual sales, indicating that the model captures the **main sales pattern effectively**.
- During periods of **steady growth and decline**, the predicted line closely aligns with the actual values, showing good model performance.
- However, there are some **deviations at peak points**, where the model tends to **overestimate or underestimate sales**, especially during high fluctuations.
- The model performs better in capturing **long-term trends** rather than short-term spikes.

Overall Interpretation

The model provides a **reasonable fit**, as it successfully captures the general direction and pattern of sales over time. However, it is slightly less accurate in predicting extreme values, suggesting that while the model is good for **trend analysis**, it may not fully capture sudden variations in sales.

8.8.3 Model Performance Metrics

The performance of the weekly model was evaluated using multiple metrics:

- RMSE (4275.12):
The model's prediction error is around **4,275 units on average**. Considering the average weekly sales (**24,671**) and maximum sales (**38,887**), this error is **moderate and acceptable**.
- MAE (3242.07):
On average, the model's predictions deviate by about **3,242 units**, indicating a **reasonable level of accuracy**.
- MAPE (23.32%):
The model has an average percentage error of about **23%**, which suggests **moderate predictive performance**.

8.8.4 Goodness of Fit

- R-squared (0.81):
The model explains approximately **81% of the variation** in weekly sales, indicating a **strong fit**.
- Adjusted R-squared:
This confirms that the model remains reliable even after considering the number of predictors.
- Pseudo R^2 (0.78):
The model explains about **78% of the variation based on deviance**, supporting its effectiveness.

Overall Interpretation

The model demonstrates **good overall performance**, with strong explanatory power and reasonable prediction accuracy. While some prediction errors exist, especially due to fluctuations in sales, the model effectively captures the **overall trend and variation** in weekly sales data.

8.9 Conclusion – Objective 3: Effect of Promotions on Sales

The analysis shows that **promotions have a strong and significant positive impact on sales**. Promoted products achieve **higher units sold and revenue** compared to non-promoted ones, which is also confirmed by the **Mann–Whitney U test (p-value = 0.0)**.

The Negative Binomial model further indicates that promotions **increase expected sales by nearly 96%**, making it the most influential factor. This effect is consistent across **regions and channels**, highlighting the overall effectiveness of promotional strategies.

However, in the weekly model, the impact of promotion becomes less significant, suggesting that **its effect is stronger in the short term**, while long-term sales are more influenced by past trends and stock availability.

Final Insight

Overall, promotions are a **key driver of sales growth** and an effective strategy to boost demand and revenue.

9. OBJECTIVE 4: CHANNEL PERFORMANCE COMPARISON

9.1 Introduction

In the FMCG industry, multiple sales channels such as Retail, E-commerce and Discount outlets are used to reach a wide range of customers. Each channel may exhibit different sales performance due to variations in customer preferences, accessibility and purchasing behaviour. Therefore, analyzing channel-wise performance is important for understanding how effectively products are distributed and sold across different platforms.

The objective of this analysis is to compare the performance of different sales channels in terms of units sold and revenue. Descriptive statistics and visualization techniques are used to examine the distribution and central tendencies of sales across channels. Furthermore, the Kruskal–Wallis test is applied to determine whether there are statistically significant differences in sales performance among the channels.

This analysis helps in identifying whether any particular channel outperforms others or whether sales are consistently distributed across all channels.

9.2 Channel-wise Descriptive Statistics

Channel	Mean Units Sold	Median Units Sold	Mean Revenue	Median Revenue
Discount	19.92	18	104.69	85.5
E-commerce	19.94	18	104.58	85.4
Retail	19.9	18	104.51	85.41

Table 9.1: Channel-wise Descriptive Statistics

The descriptive analysis shows that all three channels (**Discount, E-commerce, and Retail**) have **very similar sales performance**.

- The **average units sold** across all channels is around **19.9**, with a **median of 18**, indicating a consistent sales pattern.
- Similarly, the **average revenue (~104.5)** and **median revenue (~85.4)** are nearly the same across all channels.
- There is **no significant difference** between channels in terms of sales or revenue.

The results indicate that **all channels perform equally**, with no channel showing a clear advantage. This suggests a **balanced distribution of sales across channels**.

9.3 Average Units Sold Across Channels

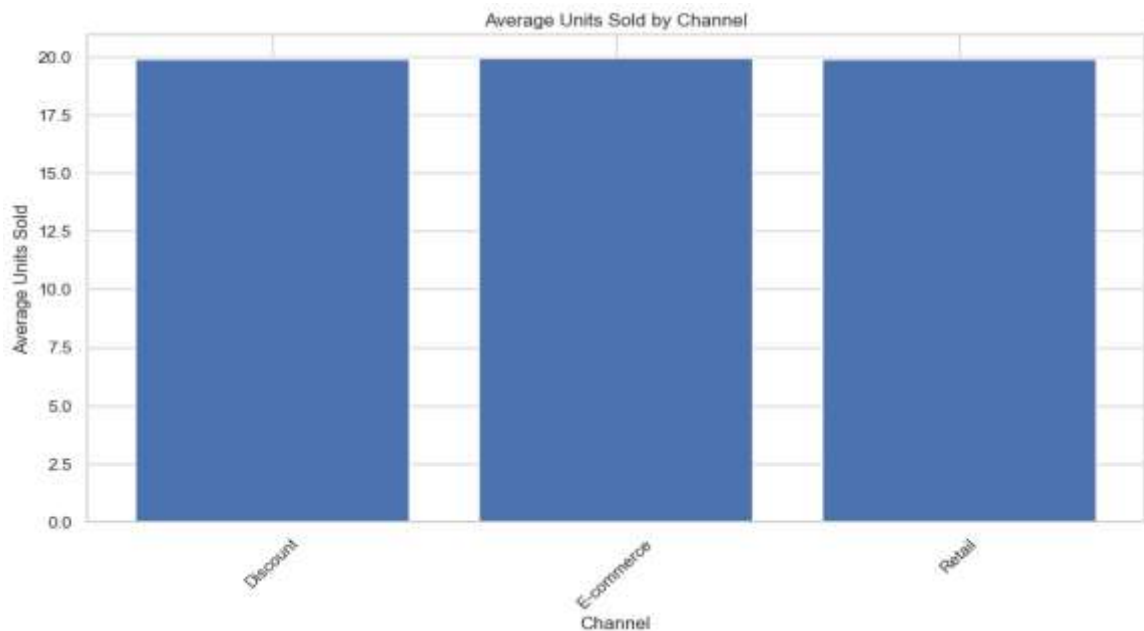


Figure 9.1: Average Units Sold Across Channels

The bar chart shows that the **average units sold across all channels (Discount, E-commerce, and Retail) are almost identical**, with values close to **20 units**.

- There is **no noticeable difference** in average sales between the channels.
- All channels perform at a **very similar level**, indicating consistent sales distribution.
- All channels perform at a **very similar level**, indicating consistent sales distribution.

This analysis further confirms that **channel type does not have a strong impact on sales performance**, and all channels contribute **equally to overall sales**.

9.4 Distribution of Units Sold Across Channels

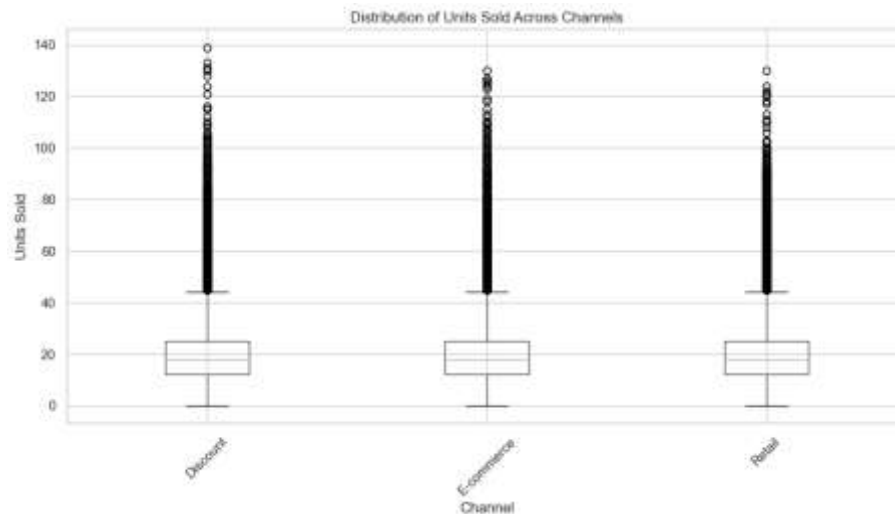


Figure 9.2: Distribution of Units Sold Across Channels

The boxplot shows that **all channels have a very similar distribution of units sold**.

- Median values are almost the same across channels.
- The spread (variability) is also similar.
- All channels have some high-value outliers, but the pattern is consistent.

This indicates that **sales behavior is similar across all channels**, with no major differences in performance.

9.5 Units Sold Comparison Across Channels

9.5.1 Hypothesis

- **Null Hypothesis (H_0):** There is **no significant difference** in units sold across different channels.
- **Alternative Hypothesis (H_1):** There is a **significant difference** in units sold across channels.

9.5.2 Test Result (Kruskal–Wallis Test)

H-statistic: 0.124

p-value: 0.9397

Interpretation

Since the **p-value (0.9397)** is much greater than **0.05**, we fail to reject the null hypothesis.

This means that there is **no statistically significant difference in units sold across the channels** (Discount, E-commerce, Retail).

The results confirm that **all channels perform similarly in terms of sales**, and channel type does not significantly impact units sold.

9.6 Effect Size Analysis

The effect size (Eta Squared) is calculated as:

$$\eta^2 = 0.00000229$$

- The effect size is **extremely close to zero**, indicating a **negligible effect** of channel on units sold.
- This means that the **variation in sales due to channel differences is almost non-existent**.
- Channel type explains **virtually none of the variation** in units sold.

The effect size analysis strongly supports previous results, confirming that **channel has no meaningful impact on sales performance**.

9.7 Revenue Comparison Across Channels

9.7.1 Hypothesis

- **Null Hypothesis (H_0):** There is **no significant difference** in revenue across channels.
- **Alternative Hypothesis (H_1):** There is a **significant difference** in revenue across channels.

9.7.2 Test Result (Kruskal–Wallis Test)

- **H-statistic:** 0.370
- **p-value:** 0.8310

Interpretation

Since the **p-value (0.8310)** is greater than **0.05**, we **fail to reject the null hypothesis**.

This indicates that there is **no statistically significant difference in revenue across channels**.

All channels generate **similar revenue**, and channel type does not significantly affect revenue performance.

9.8 Consistency Analysis (Coefficient of Variation – CV)

- **Discount:** 0.591
- **E-commerce:** 0.593
- **Retail:** 0.589

Interpretation

The CV values are **very similar across all channels**, indicating that the level of variability in sales is almost the same.

All channels show **similar consistency in sales performance**, with no channel being significantly more stable or volatile than others.

9.9 Conclusion – Objective 4: Channel Performance Comparison

The analysis shows that **all channels (Discount, E-commerce, and Retail) perform very similarly** in terms of sales and revenue. Descriptive statistics, average sales, and distribution analysis indicate **no noticeable difference** across channels.

The Kruskal–Wallis tests for both **units sold and revenue** confirm that there is **no statistically significant difference** between channels. Additionally, the **effect size is nearly zero**, showing that channel type has **no meaningful impact** on sales.

The consistency analysis (CV) also reveals that **all channels have similar variability**, indicating comparable stability in sales performance.

Final Insight

Overall, channel type does not significantly influence sales or revenue, and performance is **balanced across all channels**.

10. OBJECTIVE 5: INVENTORY & STOCK EFFICIENCY

10.1 Introduction

Efficient inventory management is a critical factor in the FMCG industry, where demand is dynamic and product availability directly influences sales performance. Maintaining an optimal balance between stock availability and sales demand is essential to avoid stockouts, minimize excess inventory, and ensure smooth supply chain operations.

The objective of this analysis is to evaluate the relationship between inventory levels and sales performance, specifically focusing on stock availability and units sold. This study examines how effectively inventory is utilized using key metrics such as stock-to-sales ratio and inventory turnover ratio. In addition, visualization techniques and correlation analysis are employed to understand the strength and nature of the relationship between stock levels and sales.

The findings from this analysis provide insights into whether higher inventory levels contribute to increased sales and help identify opportunities to improve inventory efficiency and overall operational performance.

10.2 Descriptive Analysis (Stock vs Sales)

Statistic	Stock Available	Units Sold
Count	1,90,754	1,90,754
Mean	157.7	19.92
Standard Deviation	52.73	11.77
Minimum	0	0
25th Percentile (Q1)	124	12
Median (Q2)	155	18
75th Percentile (Q3)	192	25
Maximum	405	139

Table 10.1: Descriptive Analysis (Stock vs Sales)

The summary statistics show that the **average stock available is 157.7 units**, while the **average units sold is 19.92 units**, indicating that stock levels are generally much higher than sales.

- The **median stock (155)** and **median sales (18)** suggest a consistent pattern across observations.
- The wide range in stock (**0 to 405**) and sales (**0 to 139**) indicates variability in both inventory levels and demand.
- The standard deviation shows that **stock varies more than sales**, reflecting differences in inventory management.

Overall, stock availability is significantly higher than units sold, suggesting that there may be **excess inventory in some cases**, while sales remain relatively stable.

10.3 Inventory Efficiency Metrics

10.3.1 Stock-to-Sales Ratio

- Average Stock-to-Sales Ratio = 9.18

Interpretation

This means that, on average, **stock levels are about 9 times higher than units sold.**

A high ratio indicates that **inventory is much higher compared to demand**, suggesting potential **overstocking or inefficient inventory management.**

The results imply that there is **excess stock relative to sales**, and inventory may not be optimally utilized.

10.3.2 Inventory Turnover Ratio

- Average Inventory Turnover Ratio = 0.12

Interpretation

This indicates that only **12% of the available stock is sold on average**, which is relatively low.

A low turnover ratio suggests that **inventory is moving slowly**, meaning products are not being sold quickly.

The low turnover ratio reflects **inefficient inventory utilization**, with a large portion of stock remaining unsold.

10.3.3 Stock Availability vs Sales Relationship Analysis

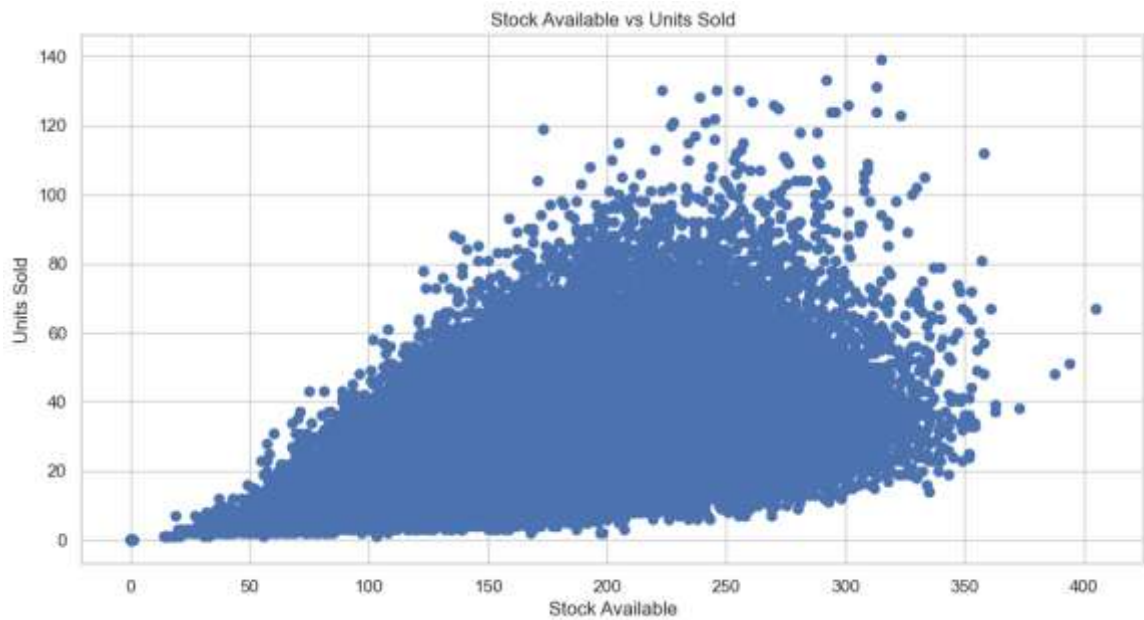


Figure 10.1: Stock Available Vs Units Sold

The scatter plot shows a **positive relationship between stock availability and units sold**.

- As **stock increases, units sold also tend to increase**, indicating that higher availability supports higher sales.
- However, the points are **widely spread**, suggesting that stock alone does not fully determine sales.
- At higher stock levels, sales do not always increase proportionally, indicating **inefficiencies or excess inventory**.

While increasing stock helps improve sales, having more stock does not guarantee higher sales, highlighting the need for better inventory optimization.

10.4 Spearman Correlation (Stock vs Sales)

10.4.1 Hypothesis

- **Null Hypothesis (H_0):** There is **no relationship** between stock availability and units sold.
- **Alternative Hypothesis (H_1):** There is a **relationship** between stock availability and units sold.

10.4.2 Test Result

- **Correlation Coefficient:** 0.638
- **p-value:** 0.0

Interpretation

Since the **p-value is less than 0.05**, we **reject the null hypothesis**.

This indicates a **statistically significant positive relationship** between stock and sales.

- As **stock increases**, **units sold also tend to increase**.
- The strength of the relationship is **moderate to strong**.

Stock availability has a **significant and meaningful impact on sales**, making it a key factor in inventory efficiency.

10.5 Conclusion – Objective 5: Inventory & Stock Efficiency

The analysis indicates that **inventory levels are significantly higher than sales**, as reflected by a high **stock-to-sales ratio (9.18)** and a low **inventory turnover ratio (0.12)**. This suggests that a large portion of stock remains unsold, pointing to **inefficient inventory utilization**.

The scatter plot and correlation analysis show a **positive and significant relationship (0.638)** between stock and sales, meaning that increasing stock can help improve sales. However, the wide variation in the data indicates that **higher stock does not always lead to proportional increases in sales**, highlighting potential overstocking.

Overall, while stock availability positively influences sales, the current inventory levels are **not optimally managed**, and there is a need to **balance stock with actual demand** to improve efficiency and reduce excess inventory.

11. OBJECTIVE 6: SALES PREDICTION (42-DAY FORECASTING)

11.1 Introduction

Accurate sales forecasting is essential in the FMCG industry to support effective inventory management, demand planning, and operational decision-making. With rapidly changing consumer demand and multiple influencing factors such as promotions, pricing, and stock availability, traditional forecasting methods may not be sufficient to capture complex patterns in sales data.

The objective of this analysis is to develop a robust machine learning model to predict product-level sales for the next 42 days. A data-driven approach is adopted using historical sales data along with advanced feature engineering techniques, including lag features, rolling statistics, momentum indicators, and seasonal components, to capture temporal trends and demand patterns.

A Light Gradient Boosting Machine (LightGBM) model is implemented to handle the large dataset and complex relationships between variables. The model is trained separately for each product category to improve prediction accuracy and account for category-specific demand behaviour.

The performance of the model is evaluated using metrics such as R-squared, Mean Absolute Percentage Error (MAPE), and Symmetric MAPE (SMAPE), and the predicted results are compared with actual sales to assess forecasting effectiveness. This analysis provides valuable insights into the model's ability to accurately predict future sales and supports better planning and decision-making.

11.2 Model Development and Train-Test Strategy

To forecast future sales, a machine learning approach was adopted using the Light Gradient Boosting Machine (LightGBM) model. LightGBM is a tree-based ensemble algorithm that is well-suited for handling large datasets and capturing complex, non-linear relationships between variables. It is computationally efficient and provides high predictive accuracy, making it appropriate for time-series forecasting tasks with multiple influencing factors.

The model was developed separately for each product category to account for differences in demand patterns across categories such as Dairy, Snacks, and Beverages. This category-wise modelling approach improves prediction accuracy by capturing category-specific trends and behaviours.

To ensure a realistic evaluation of the forecasting model, a time-based train-test split was implemented. Unlike random splitting, time-series data requires preserving the chronological order so that past observations are used to predict future values. In this study, the most recent 42 days of data were reserved as the test set, representing the forecasting horizon, while the remaining historical data was used for training. This approach simulates real-world forecasting scenarios and prevents data leakage.

Additionally, the train-test split was applied separately for each product category to ensure that category-specific models are trained and evaluated accurately, considering variations in demand patterns and data availability.

To enhance model performance, an initial LightGBM model was trained to assess feature importance, and the most relevant features were selected iteratively to minimize prediction error. This feature selection process reduced model complexity while maintaining accuracy.

The final model was trained using optimized hyperparameters such as learning rate, number of estimators, and tree depth to achieve a balance between bias and variance. The model incorporates both numerical and categorical features, enabling it to effectively learn from various aspects of the data, including historical sales, promotions, pricing, inventory levels, and seasonal patterns.

Overall, this modelling framework provides a robust and reliable approach for forecasting sales over the 42-day horizon.

11.3 Category-wise Model Performance

Category	R ² Score	SMAPE	MAPE	No. of Features
ReadyMeal	0.792	0.21	0.19	25
SnackBar	0.779	0.195	0.201	30
Yogurt	0.769	0.205	0.21	35
Milk	0.758	0.205	0.223	10
Juice	0.645	0.201	0.221	25

Table 11.1: Category-wise Model Performance

The performance of the sales forecasting model was evaluated across different product categories using key metrics such as R-squared (R^2), Mean Absolute Percentage Error (MAPE), and Symmetric MAPE (SMAPE). The results indicate that the model performs well across most categories, with some variation in prediction accuracy.

11.3.1 Overall Performance

- The **R² values range from 0.64 to 0.79**, indicating that the model explains a substantial proportion of the variation in sales.
- The **MAPE values range between ~19% and 22%**, suggesting that the model achieves reasonably accurate forecasts across categories.
- The **SMAPE values are consistent (~19%–21%)**, indicating stable prediction performance.

11.3.2 Best Performing Category

- **ReadyMeal** shows the best performance:
 - Highest R² (**0.79**) → strong explanatory power
 - Lowest MAPE (**19.0%**) → most accurate predictions
- This suggests that ReadyMeal sales patterns are **more stable and predictable**.

11.3.2 Strong Performing Categories

- **SnackBar** and **Yogurt** also perform well:
 - R² around **0.76–0.78**
 - MAPE around **20%–21%**

- These categories exhibit **moderate variability but still reliable forecasting accuracy**.

Final Conclusion

Overall, the model demonstrates strong predictive performance across categories, with higher accuracy in stable categories like ReadyMeal and slightly lower accuracy in more volatile categories such as Juice.

11.4 Best Performing SKUs (Based on MAPE)

SKU	R ² Score	SMAPE	MAPE
SN-028	0.786	0.164	0.173
RE-017	0.788	0.168	0.177
SN-010	0.785	0.184	0.18
RE-025	0.763	0.204	0.18
YO-005	0.763	0.188	0.192
YO-016	0.82	0.188	0.193
SN-030	0.748	0.213	0.193
RE-007	0.833	0.212	0.196
MI-006	0.812	0.174	0.197
RE-015	0.777	0.233	0.198

Table 11.2: Best Performing SKUs (Based on MAPE)

To evaluate the model's performance at a more granular level, SKU-wise analysis was conducted using key metrics such as R-squared (R^2), MAPE, and SMAPE. The results highlight the top-performing SKUs with the lowest prediction errors.

11.4.1 Overall Observation

- The **MAPE values for the best SKUs range between ~17% and 20%**, indicating **high forecasting accuracy** at the individual product level.
- The **R² values are consistently high (0.75 to 0.83)**, showing that the model explains a significant portion of the variation in sales for these SKUs.

11.4.2 Top Performing SKUs

- **SN-028** shows the best performance:
 - Lowest MAPE (**17.3%**)

- Low SMAPE (**16.4%**)
- This indicates **very accurate and stable predictions**.
- Other strong performers include:
 - RE-017, SN-010, RE-025
 - MAPE around **17%–18%**
 - High R^2 (~ 0.76 – 0.79)

11.4.3 High Accuracy with Strong Fit

- Some SKUs such as **YO-016** and **RE-007** have:
 - Very high R^2 (**0.82–0.83**)
 - Slightly higher MAPE (~ 19 – 20%)
- This suggests the model captures overall trends well, but minor fluctuations still exist.

11.4.4 Key Insight

- These top-performing SKUs likely have:
 - Stable demand patterns
 - Consistent sales behavior
 - Less volatility from promotions or external factors

Final Conclusion

The model demonstrates strong predictive accuracy for several SKUs, particularly those with stable and consistent demand patterns, achieving low error rates and high explanatory power.

11.5 Worst Performing SKUs (Based on MAPE)

SKU	R ² Score	SMAPE	MAPE
MI-008	0.61	0.223	0.271
MI-011	0.717	0.215	0.254
SN-013	0.726	0.196	0.236
YO-014	0.736	0.204	0.234
MI-022	0.755	0.191	0.227
SN-027	0.746	0.197	0.224
YO-001	0.745	0.199	0.224
JU-021	0.645	0.201	0.221
YO-003	0.725	0.212	0.215
MI-002	0.713	0.218	0.213

Table 11.3: Worst Performing SKUs (Based on MAPE)

In addition to the best-performing products, SKU-level analysis was conducted to identify items where the model shows relatively lower predictive accuracy. These SKUs exhibit higher error values, indicating challenges in forecasting their demand patterns.

11.5.1 Overall Observation

- The **MAPE values for the worst-performing SKUs range from ~21% to 27%**, indicating comparatively higher prediction errors.
- The **R² values range from 0.61 to 0.75**, suggesting moderate model performance but weaker explanatory power compared to top-performing SKUs.

11.5.2 Lowest Performing SKU

- **MI-008** shows the weakest performance:
 - Highest MAPE (**27.1%**)
 - Lowest R² (**0.61**)
- This indicates **high variability and difficulty in capturing demand patterns**.

11.5.3 Other Underperforming SKUs

- MI-011, SN-013, YO-014
 - MAPE between **23%–25%**
 - Moderate R² (**~0.71–0.73**)

- These SKUs show **inconsistent prediction accuracy**, likely due to fluctuating demand.

11.5.4 Moderate Errors with Acceptable Fit

- SKUs like **MI-022, SN-027, YO-001**
 - MAPE around **22%–23%**
 - R^2 around **0.74–0.75**
- The model performs reasonably well but still faces **minor prediction deviations**.

11.5.5 Key Reasons for Lower Performance

These SKUs may exhibit:

- High demand variability
- Frequent promotions or irregular sales spikes
- Stock fluctuations or supply constraints
- External influencing factors not captured in the model

Final Conclusion

The model shows relatively lower accuracy for certain SKUs with volatile and irregular demand patterns, highlighting areas where forecasting can be further improved.

11.6 Weekly Actual vs Predicted Sales

11.6.1 Yogurt:



Figure 11.1: Yogurt – Weekly Prediction Vs Actual

For the **Yogurt** category, the predicted sales closely match the actual sales across most weeks. The model successfully captures both the rising and declining trend in sales, with only minor deviations in peak weeks. This indicates that the forecasting model performs strongly for Yogurt and is able to represent its sales dynamics effectively.

11.6.2 ReadyMeal:

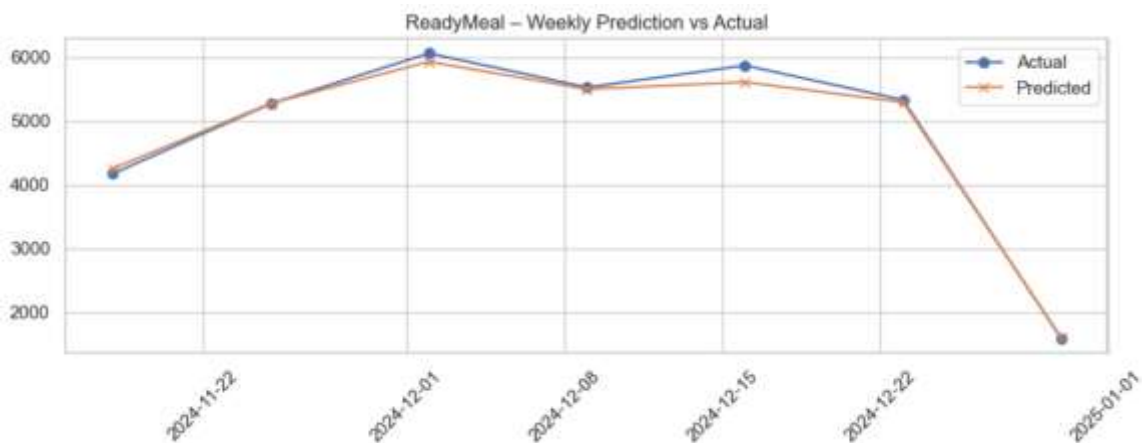


Figure 11.2: ReadyMeal – Weekly Prediction Vs Actual

In the **ReadyMeal** category, the predicted values are also highly aligned with the actual sales pattern. The model captures the general weekly movement well, although slight underestimation can be observed during certain peak periods. Overall, the closeness of the two lines suggests reliable forecasting performance for this category.

11.6.3 Milk:

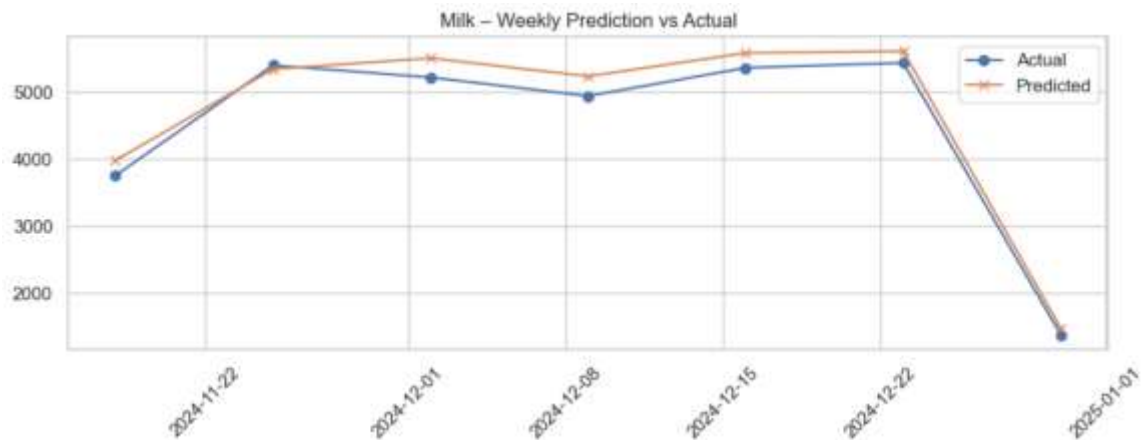


Figure 11.3: Milk – Weekly Prediction Vs Actual

For **Milk**, the model follows the actual trend reasonably well, with predictions remaining close to the observed values throughout the forecast period. However, compared to Yogurt and ReadyMeal, slightly larger deviations are visible in some weeks, indicating moderate forecasting error. Even so, the model still captures the broad trend effectively.

Overall Interpretation

Overall, the weekly comparison between actual and predicted sales demonstrates that the model is effective in capturing category-level sales trends over the 42-day forecasting period. The predictions remain close to the actual values across the selected categories, suggesting that the model can be used as a reliable tool for short-term sales forecasting. Minor deviations observed in some peak weeks indicate that highly fluctuating demand remains more challenging to predict precisely.

11.7 Conclusion – Objective 6: Sales Prediction (42-Day Forecasting)

The LightGBM model successfully achieved accurate sales forecasting over a 42-day horizon by capturing key factors such as historical demand, promotions, pricing, inventory, and seasonality. The model demonstrated good predictive performance across product categories, with higher accuracy for stable categories like ReadyMeal and SnackBar, while slightly higher errors were observed in more variable categories such as Juice and Milk.

The comparison of actual and predicted sales shows that the model effectively follows overall sales trends, making it suitable for short-term forecasting. Additionally, SKU-level analysis indicates better performance for products with consistent demand, whereas highly fluctuating items remain more challenging to predict.

Overall, the model provides a reliable and practical solution for sales forecasting, supporting improved inventory planning and business decision-making.

12. FINAL PROJECT CONCLUSION

This study presents a comprehensive analysis of sales patterns by evaluating the impact of key factors such as pricing, promotions, sales channels, and inventory, along with the development of a robust forecasting model. The results clearly indicate that **promotions and stock availability are the most significant drivers of sales**, with promotional activities leading to a substantial increase in demand. On the other hand, **price and channel variations show minimal influence on sales performance**, suggesting consistent behaviour across different channels and limited price sensitivity in the dataset.

The analysis further reveals that **sales follow strong temporal patterns**, where past sales significantly influence future demand, highlighting the importance of incorporating lag-based features in modelling. The use of the **Negative Binomial model** effectively addresses the count nature of the data and overdispersion, confirming the statistical significance of key variables.

In addition, the forecasting model demonstrates **strong predictive performance across product categories**, successfully capturing overall sales trends with reasonable accuracy. Higher accuracy is observed for products with stable demand, while slightly lower performance is seen for highly variable items.

Overall, the study concludes that **strategic use of promotions, efficient inventory management, and leveraging historical sales patterns are essential for improving sales performance and forecasting accuracy**, thereby supporting more informed and effective business decision-making.

13. RECOMMENDATIONS

Based on the analysis and modeling results, the following recommendations are suggested to improve sales performance and decision-making:

- **Leverage Promotions Strategically:**
Since promotions have a strong positive impact on sales, businesses should invest in well-planned promotional campaigns to boost demand. Targeted and timely promotions can significantly enhance sales performance.
- **Optimize Inventory Management:**
Stock availability is a key driver of sales. Maintaining optimal inventory levels can help prevent stockouts and lost sales opportunities, while also avoiding overstocking.
- **Focus Less on Price Adjustments:**
As price shows minimal impact on sales in this analysis, frequent price changes may not be an effective strategy. Instead, emphasis should be placed on promotions and availability.
- **Utilize Demand Forecasting Models:**
The forecasting model demonstrates reliable performance and should be used for **short-term planning**, including inventory allocation, production planning, and demand estimation.
- **Incorporate Historical Sales Trends:**
Since past sales strongly influence future demand, businesses should leverage historical data and lag-based insights for more accurate forecasting and planning.
- **Improve Performance for Volatile Products:**
Products with high demand variability should be monitored closely. Additional factors such as seasonality, external influences, or customer behavior can be incorporated to improve prediction accuracy.
- **Ensure Consistent Channel Strategy:**
As no major differences were observed across channels, maintaining a balanced and consistent strategy across all sales channels is recommended.

14. REFERENCES

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15. APPENDIX

Appendix A: Dataset Source

<https://www.kaggle.com/datasets/beatafaron/fmcg-daily-sales-data-to-2022-2024>

Appendix B: For full Python Code, R Script, HTML File and report file visit below link

<https://github.com/Tushar-ai28/Sem-4-Project>

